

CS355 Mini-Project

Team No.: 11

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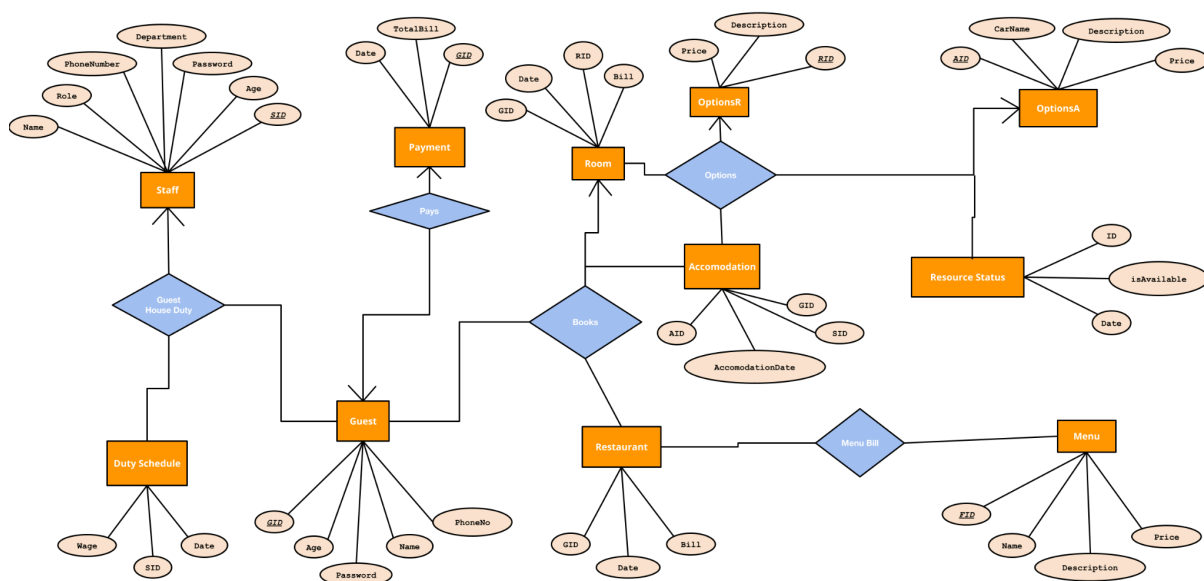
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Task A: Guest House Related Service

We were required to create guest house module in our mini project which was capable of doing the following tasks: -

- Monthly bookings for the guest house in different categories
- The total monthly expenditure for the guest house
- Generation of bills
- Availability of room
- Monthly food billing

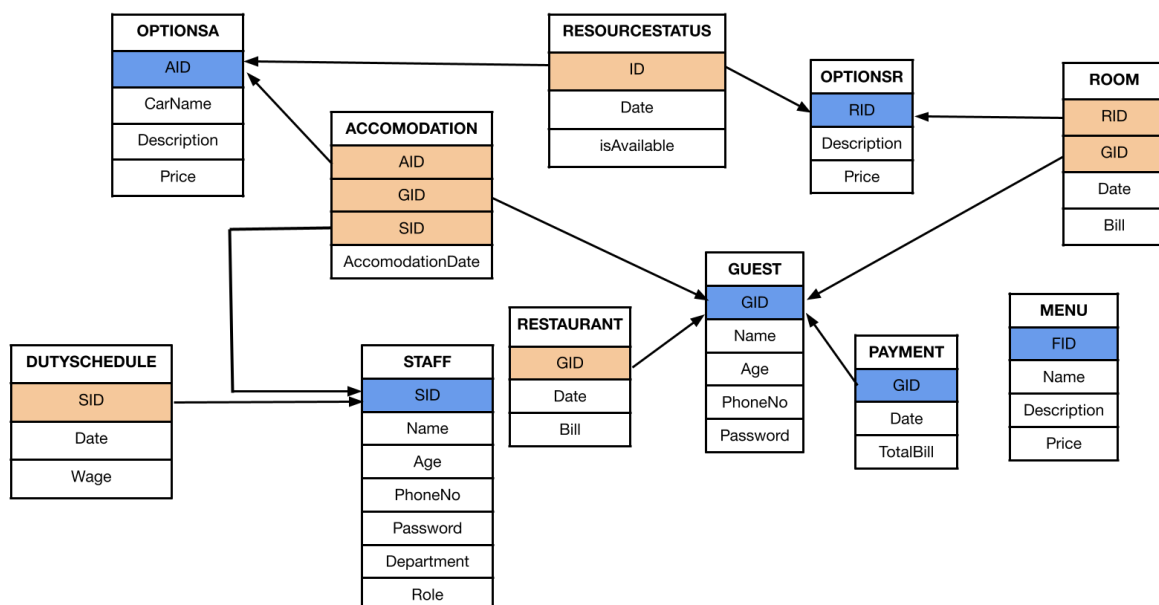
E-R Model



The relationships are described as follows:

- Guest House Duty - Relates that due to booking of a guest, a staff member is assigned a duty. Only one staff member can be assigned duty for a booking by a guest, so it is a one-to-many relationship with staff at one end.
- Pays - Guests make a payment for bookings.
- Books - Guest books accommodation, rooms and food.
- Menu bill - Food order creates a bill for the restaurant.
- Options - Room and accommodation bookings have different options to choose from which depend upon the status if they are available or not.

Relational Model



Primary key

Primary key is given in blue.

AID, SID, GID, RID, FID are the primary keys for their tables.

Foreign key

Foreign key is given in orange.

Staff can be assigned a duty only when it is in the staff table.

A guest can order food and make payment only when it is in the guest table.

A room can be booked only when the guest is in guest table and room is in optionsr table.

The Tables are optimised as we can see.

They are in 1NF as all values in all tables are atomic.

They are also optimised to be in 2NF form.

They are in 3NF form as there is no transitive partial dependency for non-prime attributes in the tables

They are also in BCNF as for every attribute it is only dependent on the primary key and Hence all LHS of Functional dependency will be SuperKey.

Queries

Create database

```
DROP DATABASE IF EXISTS guesthouse;

CREATE DATABASE guesthouse;

USE guesthouse;
```

Create tables

- Guest - Stores data about the guests which register on the site and use the portal to do the bookings.

```
CREATE TABLE `guest` (
  `gid` varchar(100) NOT NULL,
  `name` varchar(100) NOT NULL,
  `age` int(11) NOT NULL,
  `phoneNo` int(11) NOT NULL,
  `password` varchar(100) NOT NULL
```

```
);  
  
ALTER TABLE `guest`  
  ADD PRIMARY KEY (`gid`);
```

- Accommodation - Stores data about taxi bookings like ids of guest, taxi, driver and date.

```
CREATE TABLE `Accomodation` (  
  `AID` varchar(100) DEFAULT NULL,  
  `GID` varchar(100) NOT NULL,  
  `SID` varchar(100) DEFAULT NULL,  
  `accomodationDate` date DEFAULT NULL  
);
```

- Dutyschedule - Stores data about duty schedule of the staff.

```
CREATE TABLE `dutySchedule` (  
  `sid` varchar(100) NOT NULL,  
  `date` date NOT NULL,  
  `wage` int(11) NOT NULL  
);
```

- Menu - Stores data about menu items for the food booking.

```
CREATE TABLE `menu` (  
  `fid` varchar(100) NOT NULL,  
  `name` varchar(100) DEFAULT NULL,  
  `description` varchar(100) DEFAULT NULL,  
  `price` int(11) DEFAULT NULL  
);  
  
ALTER TABLE `menu`  
  ADD PRIMARY KEY (`fid`);
```

- Payment - Stores bill due for a guest.

```
CREATE TABLE `payment` (  
  `GID` varchar(100) NOT NULL,  
  `date` date DEFAULT NULL,  
  `totalBill` int(11) DEFAULT NULL  
);
```

```
ALTER TABLE `payment`  
  ADD PRIMARY KEY (`GID`);
```

- resourceStatus - Stores the availability of a resource.

```
CREATE TABLE `resourceStatus` (  
  `ID` varchar(100) NOT NULL,  
  `Date` date DEFAULT NULL,  
  `isAvailable` int(11) DEFAULT NULL  
);
```

- restaurant - Stores the data about food orders of a guest.

```
CREATE TABLE `restaurant` (  
  `gid` varchar(100) NOT NULL,  
  `date` date NOT NULL,  
  `bill` int(11) NOT NULL  
);
```

- Room - Stores data about room booking of a guest.

```
CREATE TABLE `Room` (  
  `RID` varchar(100) DEFAULT NULL,  
  `GID` varchar(100) NOT NULL,  
  `date` date NOT NULL,  
  `bill` int(11) NOT NULL  
);
```

- Staff - Stores all the data about the staff.

```
CREATE TABLE `staff` (  
  `SID` varchar(100) NOT NULL,  
  `Name` varchar(100) DEFAULT NULL,  
  `Age` int(11) DEFAULT NULL,  
  `PhoneNo` int(11) DEFAULT NULL,  
  `Department` varchar(100) DEFAULT NULL,  
  `Role` varchar(100) DEFAULT NULL,  
  `password` varchar(100) NOT NULL  
);  
  
ALTER TABLE `staff`  
  ADD PRIMARY KEY (`SID`);
```

- OptionsA - Stores data of the transport services.

```
CREATE TABLE `optionsA` (  
  `AID` varchar(100) NOT NULL,  
  `carName` varchar(100) DEFAULT NULL,  
  `description` varchar(500) DEFAULT NULL,  
  `price` int(11) DEFAULT NULL  
);  
  
ALTER TABLE `optionsA`  
  ADD PRIMARY KEY (`AID`);
```

- OptionsR - Stores data about the rooms.

```
CREATE TABLE `optionsR` (  
  `RID` varchar(100) NOT NULL,  
  `description` varchar(500) DEFAULT NULL,  
  `price` int(11) DEFAULT NULL  
);  
  
ALTER TABLE `optionsR`  
  ADD PRIMARY KEY (`RID`);
```

Populating tables

```
INSERT INTO `guest` (`gid`, `name`, `age`, `phoneNo`, `password`) VALUES  
( '1901CS01', 'Akash', 20, 85278, '1'),  
( '1901CS02', 'Bob', 20, 12345, '1'),  
( '1901CS03', 'Chirag', 28, 12345, '1'),  
( '1901EE01', 'Dheeraj', 20, 85245, '1'),  
( '1901EE02', 'Eyesha', 80, 98157, '1'),  
( '1901EE03', 'Faraz', 40, 87491, '1'),  
( '1901ME01', 'Ganesh', 80, 98157, '1'),  
( '1901ME02', 'Himesh', 80, 98157, '1'),  
( '1901ME03', 'India', 80, 98157, '1');
```

```
INSERT INTO `menu` (`fid`, `name`, `description`, `price`) VALUES  
( 'F1', 'Pasta', 'Spaghetti with rich tomato aroma and cheese', 300),  
( 'F10', 'Dal Makhni', 'Blend of four lentils in smooth flavoured curry', 200),  
( 'F11', 'Chicken Korma', 'Boneless chicken in curry of cashew and mild  
spices', 200),
```

```
( 'F2', 'Vegetarian Tandoori Platter', 'Melenge of exotic vegetable
preparation', 395),
( 'F3', 'Extra Cold Coffee', 'Cold Coffee served with ice-cream', 100),
( 'F4', 'Thandai', 'Traditional Drink with Almond, Saffron, Cardamom in milk',
150),
( 'F5', 'Egg Sandwich', 'Fresh omelette with buttered bread and chopped
vegetables', 40),
( 'F6', 'Cheese Sandwich', 'Cheese is sliced onto bread with chopped
vegetables', 30),
( 'F7', 'Coffee', 'Freshly Grounded Hot Coffee', 20),
( 'F8', 'Pancakes', 'Maple Syruped Pancake', 50),
( 'F9', 'Butter Chicken', 'Sweet curry with fresh chicken', 200);
```

```
INSERT INTO `optionsA` (`AID`, `carName`, `description`, `price`) VALUES
( 'A1', 'Sedan', 'Car Hire For Outstation', 2000),
( 'A2', 'SUV', 'Car Hire For Outstation', 3000),
( 'A3', 'Jeep', 'Car Hire For Outstation', 3000),
( 'A4', 'Indigo', 'Car Hire For Outstation', 1500),
( 'A5', 'Taxi', 'Car to railway/bus/airport', 1000),
( 'A6', 'i10', 'Car to railway/bus/airport', 1500);
```

```
INSERT INTO `optionsR` (`RID`, `description`, `price`) VALUES
( 'R1', 'Single Room', 1000),
( 'R2', 'Double Room', 1500),
( 'R3', 'King Room', 2000),
( 'R4', 'Queen Room', 2000),
( 'R5', 'Twin Room', 2000),
( 'R6', 'Executive Suite', 2500),
( 'R7', 'Room for Extended Stay', 3000);
```

```
INSERT INTO `staff` (`SID`, `Name`, `Age`, `PhoneNo`, `Department`, `Role`,
`password`) VALUES
( 'S1', 'Abhay', 25, 14134, 'Restaurant', 'Chef', '1'),
( 'S2', 'Bijnoy', 23, 34313, 'Restaurant', 'Restaurant General Manager', '1'),
( 'S3', 'Chaman', 23, 52513, 'Hotel', 'Room Service', '1'),
( 'S4', 'Dilraj', 32, 46242, 'Accommodation', 'Driver', '1'),
( 'S5', 'Easter', 32, 12133, 'Hotel', 'Manager', '1'),
( 'S6', 'Farah', 25, 12341, 'Hotel', 'House Cleaning', '1');
```

After inserting more data through web-interface, count of data in tables is as follows:

QUERY	OUTPUT
SELECT COUNT(*) FROM accomodation;	5
SELECT COUNT(*) FROM dutyschedule;	8
SELECT COUNT(*) FROM guest;	8
SELECT COUNT(*) FROM menu;	11
SELECT COUNT(*) FROM optionsa;	6
SELECT COUNT(*) FROM optionsr;	7
SELECT COUNT(*) FROM payment;	5
SELECT COUNT(*) FROM resourcestatus;	8

SELECT COUNT(*) FROM restaurant;	3
SELECT COUNT(*) FROM room;	6
SELECT COUNT(*) FROM staff;	6

Important code snippets and queries

- Login for admin

```
// Admin login
if(isset($_POST['admin_login'])) {
    $pass = mysqli_real_escape_string($db, $_POST['password']);

    if($pass == '123') {
        $_SESSION['admin'] = "good";
        header('location: admin.php');
    }
    else {
        array_push($errors, "Wrong Admin Password");
    }
}
```


- Registration for guest

```
// Guest register
if (isset($_POST['reg_user'])) {

    // storing the user input in variables
    $name = mysqli_real_escape_string($db, $_POST['name']);
    $age = mysqli_real_escape_string($db, $_POST['age']);
    $mobileNumber = mysqli_real_escape_string($db, $_POST['mobileNo']);
    $username = mysqli_real_escape_string($db, $_POST['username']);
    $password_1 = mysqli_real_escape_string($db, $_POST['password_1']);
    $password_2 = mysqli_real_escape_string($db, $_POST['password_2']);

    // form validation
    if (empty($username)) {
        array_push($errors, "Guest ID is required");
    }
    if (empty($name)) {
        array_push($errors, "Guest Name is required");
    }
    if (empty($mobileNumber)) {
        array_push($errors, "Phone Number is required");
    }
    if (empty($age)) {
        array_push($errors, "Age is required");
    }
    if (empty($password_1) || empty($password_2)) {
        array_push($errors, "Password is required");
    }
    if ($password_1 != $password_2) {
        array_push($errors, "Passwords do not match");
    }

    $user_check_query = "SELECT * FROM guest WHERE GID = '$username'";
    $result = mysqli_query($db, $user_check_query);
    $user = mysqli_fetch_assoc($result);

    if ($user) {
        array_push($errors, "Guest ID already exists, please proceed to
login");
    }
}
```

```

        // if no errors, enter data into database
        if (count($errors) == 0) {
            $query = "INSERT INTO guest (GID, name, age, phoneNo, password)
VALUES ('$username', '$name', '$age', '$mobileNumber', '$password_1') ";
            $check = mysqli_query($db, $query);

            $_SESSION['username'] = $username;
            $_SESSION['Guestname'] = $name;
            $_SESSION['GuestmobileNumber'] = $mobileNumber;
            $_SESSION['Guestage'] = $age;
            $_SESSION['GuestPassword'] = $password_1;

            $_SESSION['success'] = "You are now Logged In";
            header('location: GuestInfo.php');
        }
    }
}

```

- Login for guest

```

// Guest login
if (isset($_POST['login_user'])) {

    // storing input from user in variables
    $username = mysqli_real_escape_string($db, $_POST['gid']);
    $password = mysqli_real_escape_string($db, $_POST['password']);

    // neither of fields should be empty
    if (empty($username)) {
        array_push($errors, "Guest ID is required.");
    }
    if (empty($password)) {
        array_push($errors, "Password is required.");
    }

    if (count($errors) == 0) {
        $query = "SELECT * FROM guest WHERE GID = '$username' AND password
= '$password' ";
        $results = mysqli_query($db, $query);

        if (mysqli_num_rows($results) == 1) {
            $row = mysqli_fetch_assoc($results);

```

```

        // store guest data in SESSION for further use
        $_SESSION['username'] = $username;
        $_SESSION['Guestname'] = $row["name"];
        $_SESSION['GuestmobileNumber'] = $row["phoneNo"];
        $_SESSION['Guestage'] = $row["age"];
        $_SESSION['GuestPassword'] = $row["password"];

        $_SESSION['success'] = "Successfully logged in!";
        header('location: GuestInfo.php');
    } else {
        array_push($errors, "Wrong username/password combination.");
    }
}
}

```

- Login for staff

```

// Staff login
if (isset($_POST['staff_login'])) {

    // storing input from user in variables
    $username = mysqli_real_escape_string($db, $_POST['sid']);
    $password = mysqli_real_escape_string($db, $_POST['sidpassword']);

    //neither of fields should be empty
    if (empty($username)) array_push($errors, "Employee ID is required");
    if (empty($password)) array_push($errors, "Password is required");

    if (count($errors) == 0) { // if no errors then proceed

        $query = "SELECT * FROM staff WHERE sid = '$username' AND password
= '$password'";
        $results = mysqli_query($db, $query);

        if (mysqli_num_rows($results) == 1) {

            // query to update
            $row_ = mysqli_fetch_assoc($results); // stores details about
the same employee after update

```

```

        $_SESSION['Susername'] = $username;
        $_SESSION['Spassword'] = $password;
        $_SESSION['success'] = "You are now logged in";
        header('location: staffInfo.php');
    }
    else {
        array_push($errors, "Wrong username/password combination");
    }
}
}

```

- Getting all the booking details of a guest like room id, car, driver, etc.

```

$acc = mysqli_query($db, "SELECT * FROM Accomodation WHERE gid = '$gid'");
$res = mysqli_query($db, "SELECT * FROM restaurant WHERE gid = '$gid'");
$hotel = mysqli_query($db, "SELECT * FROM Room WHERE gid = '$gid'");
$payment = mysqli_query($db, "SELECT * FROM payment WHERE gid = '$gid'");
$query = mysqli_fetch_assoc(mysqli_query($db, "SELECT SUM(bill) AS A FROM restaurant WHERE gid = '$gid'"));
$totalResBill = $query['A'];
$room = mysqli_fetch_assoc(mysqli_query($db, "SELECT * FROM optionsR WHERE rid = '$rid' "));
$car = mysqli_fetch_assoc(mysqli_query($db, "SELECT * FROM optionsA WHERE aid = '$aid' "));
$driver = mysqli_fetch_assoc(mysqli_query($db, "SELECT * FROM staff WHERE sid = '$sid' "));

```

- Getting wage earned by a staff

```

$query = "select * from dutySchecule where sid = '$sid'";
$query_ = "select sum(wage) as A from dutySchecule where sid = '$sid'";

```

- Getting information about a staff

```

$query = "select * from staff where sid = '$sid' and password = '$password' ";
$check = mysqli_query($db, $query);
$row = mysqli_fetch_assoc($check);

$name = $row['Name'];
$age = $row['Age'];
$phoneNo = $row['PhoneNo'];

```

```
$dep = $row['Department'];  
$role = $row['Role'];
```

- Checking the availability of rooms and transport

```
// Room and transportation options  
if (isset($_POST['findOptions'])) {  
    date_default_timezone_set('Asia/Kolkata');  
  
    $timezone = date('Y-m-d');  
  
    $date1 = mysqli_real_escape_string($db, $_POST['date1']);  
  
    $date2;  
  
    if($_SESSION['date1'] === "Pick-up Date")  
        $date2 = $date1;  
    else  
        $date2 = mysqli_real_escape_string($db, $_POST['date2']);  
  
    $_SESSION['orderDate'] = $date2;  
  
    if ($date2 < $date1) {  
        array_push($errors, "Dates not valid");  
    } elseif ($date1 < $timezone) {  
        array_push($errors, "Please enter date after or today");  
    } elseif ($_SESSION['date1'] == "Check-in Date") {  
        $query1 = "Select date from resourceStatus where id = 'R1' and date  
between '$date1' AND '$date2'";  
        $query2 = "Select date from resourceStatus where id = 'R2' and date  
between '$date1' AND '$date2'";  
        $query3 = "Select date from resourceStatus where id = 'R3' and date  
between '$date1' AND '$date2'";  
        $query4 = "Select date from resourceStatus where id = 'R4' and date  
between '$date1' AND '$date2'";  
        $query5 = "Select date from resourceStatus where id = 'R5' and date  
between '$date1' AND '$date2'";  
        $query6 = "Select date from resourceStatus where id = 'R6' and date  
between '$date1' AND '$date2'";  
        $query7 = "Select date from resourceStatus where id = 'R7' and date  
between '$date1' AND '$date2'";
```

```
$results1 = mysqli_fetch_assoc(mysqli_query($db, $query1));
$results2 = mysqli_fetch_assoc(mysqli_query($db, $query2));
$results3 = mysqli_fetch_assoc(mysqli_query($db, $query3));
$results4 = mysqli_fetch_assoc(mysqli_query($db, $query4));
$results5 = mysqli_fetch_assoc(mysqli_query($db, $query5));
$results6 = mysqli_fetch_assoc(mysqli_query($db, $query6));
$results7 = mysqli_fetch_assoc(mysqli_query($db, $query7));

$_SESSION['A'] = "Available";
$_SESSION['B'] = "Available";
$_SESSION['C'] = "Available";
$_SESSION['D'] = "Available";
$_SESSION['E'] = "Available";
$_SESSION['F'] = "Available";
$_SESSION['G'] = "Available";

if (!empty($results1)) {
    $_SESSION['A'] = "Not Available";
}

if (!empty($results2)) {
    $_SESSION['B'] = "Not Available";
}

if (!empty($results3)) {
    $_SESSION['C'] = "Not Available";
}

if (!empty($results4)) {
    $_SESSION['D'] = "Not Available";
}

if (!empty($results5)) {
    $_SESSION['E'] = "Not Available";
}

if (!empty($results6)) {
    $_SESSION['F'] = "Not Available";
}
```

```

        if (!empty($results7)) {
            $_SESSION['G'] = "Not Available";
        }

        $_SESSION['startDateRoom'] = $date1;
        $_SESSION['endDateRoom'] = $date2;

        $_SESSION['success'] = "You are now logged in";
        header('location: room.php');
    } else {
        if ($date1 == $date2) {
            $query1 = "Select date from resourceStatus where id ='A1' and
date between '$date1' AND '$date2'";
            $query2 = "Select date from resourceStatus where id ='A2' and
date between '$date1' AND '$date2'";
            $query3 = "Select date from resourceStatus where id ='A3' and
date between '$date1' AND '$date2'";
            $query4 = "Select date from resourceStatus where id ='A4' and
date between '$date1' AND '$date2'";
            $query5 = "Select date from resourceStatus where id ='A5' and
date between '$date1' AND '$date2'";
            $query6 = "Select date from resourceStatus where id ='A6' and
date between '$date1' AND '$date2'";

            $results1 = mysqli_fetch_assoc(mysqli_query($db, $query1));
            $results2 = mysqli_fetch_assoc(mysqli_query($db, $query2));
            $results3 = mysqli_fetch_assoc(mysqli_query($db, $query3));
            $results4 = mysqli_fetch_assoc(mysqli_query($db, $query4));
            $results5 = mysqli_fetch_assoc(mysqli_query($db, $query5));
            $results6 = mysqli_fetch_assoc(mysqli_query($db, $query6));

            $_SESSION['A'] = "Available";
            $_SESSION['B'] = "Available";
            $_SESSION['C'] = "Available";
            $_SESSION['D'] = "Available";
            $_SESSION['E'] = "Available";
            $_SESSION['F'] = "Available";

            if (!empty($results1)) {
                $_SESSION['A'] = "Not Available";
            }

```

```

        if (!empty($results2)) {
            $_SESSION['B'] = "Not Available";
        }

        if (!empty($results3)) {
            $_SESSION['C'] = "Not Available";
        }

        if (!empty($results4)) {
            $_SESSION['D'] = "Not Available";
        }

        if (!empty($results5)) {
            $_SESSION['E'] = "Not Available";
        }

        if (!empty($results6)) {
            $_SESSION['F'] = "Not Available";
        }

        $_SESSION['startDateACC'] = $date1;

        $_SESSION['success'] = "You are now logged in";
        header('location: transportation.php');
    } else {
        array_push($errors, "Transportation Booking can be done only
for one day");
    }
}
}

```

- Updating the tables when a booking is made

```

// Post-payment updation of tables
if (isset($_POST['payIt'])) {
    $gid = $_SESSION['username'];

    $name = mysqli_real_escape_string($db, $_POST['name']);
    $phoneNo = mysqli_real_escape_string($db, $_POST['phoneNo']);
    $username = mysqli_real_escape_string($db, $_POST['username']);

```

```

$cardname = mysqli_real_escape_string($db, $_POST['cardname']);
$cardnumber = mysqli_real_escape_string($db, $_POST['cardnumber']);
$expmonth = mysqli_real_escape_string($db, $_POST['expmonth']);
$expyear = mysqli_real_escape_string($db, $_POST['expyear']);
$cvv = mysqli_real_escape_string($db, $_POST['cvv']);

$query = "SELECT * FROM guest WHERE gid = '$gid'";
$row = mysqli_fetch_assoc(mysqli_query($db, $query));

if ($username !== $gid) {
    array_push($errors, "Guest ID does not match");
}

date_default_timezone_set('Asia/Kolkata');
$timezone = date('Y-m-d');

$year = date('Y', strtotime($timezone));
$month = date('F', strtotime($timezone));

if ($expmonth !== "January" &&
    $expmonth !== "February" &&
    $expmonth !== "March" &&
    $expmonth !== "April" &&
    $expmonth !== "May" &&
    $expmonth !== "June" &&
    $expmonth !== "July" &&
    $expmonth !== "August" &&
    $expmonth !== "September" &&
    $expmonth !== "October" &&
    $expmonth !== "November" &&
    $expmonth !== "December") {
    array_push($errors, "Please enter valid Expiry Month");
}

if ($expyear < $year) {
    array_push($errors, "Card has expired");
} elseif ($expyear === $year && $month < $expmonth) {
    array_push($errors, "Card has expired");
}

if (count($errors)==0) {

```

```

        $bill = $_SESSION['bill'];
        $check = mysqli_query($db, "SELECT * FROM payment WHERE gid =
'$gid' ");

        if (mysqli_num_rows($check) == 1) {
            $update_ = "UPDATE payment SET totalbill = totalbill +
'$bill', date = '$timezone' WHERE gid = '$gid' ";
            $row = mysqli_query($db, $update_);
        } else {
            $insert = "INSERT INTO payment (GID, date, totalBill) VALUES
('$gid', '$timezone', '$bill')";
            $check = mysqli_query($db, $insert);
        }

        if($_SESSION['what'] === "restaurant"){
            $insert = "INSERT INTO restaurant (gid, date, bill) VALUES
('$gid', '$timezone', '$bill')";
            mysqli_query($db, $insert);

            $staff = "INSERT INTO dutySchecule (sid, date, wage) VALUES
('S3', '$timezone', 50)";
            mysqli_query($db, $staff);

            $staff_ = "INSERT INTO dutySchecule (sid, date, wage) VALUES
('S1', '$timezone', 200)";
            mysqli_query($db, $staff_);
        }
        else if($_SESSION['what'] == "room") {
            $rid = $_SESSION['Roomrid'];
            $date1 = $_SESSION['startDateRoom'];
            $date2 = $_SESSION['endDateRoom'];

            $insert = "INSERT INTO Room (rid, gid, date, bill) VALUES
('$rid', '$gid', '$timezone', '$bill')";
            $B = mysqli_query($db, $insert);

            $diff = abs(strtotime($date2) - strtotime($date1));
            $years = floor($diff / (365*60*60*24));
            $months = floor(($diff - $years * 365*60*60*24) /
(30*60*60*24));

```

```

        $days = floor(($diff - $years * 365*60*60*24 -
$months*30*60*60*24)/ (60*60*24));

        while($days--) {
            echo $date1;
            echo " ";
            $insert_ = "INSERT INTO resourceStatus (id, date,
isAvailable) VALUES ('$rid','$date1', 0)";
            $date1 = date('Y-m-d', strtotime("+1 day",
strtotime($date1)));
            $A = mysqli_query($db, $insert_ );
        }

        $staff = "INSERT INTO dutySchecule (sid, date, wage) VALUES
('S6', '$timezone', 100)";
        mysqli_query($db, $staff);
    }
    else if($_SESSION['what'] == "accomodation") {
        $order = $_SESSION['Cardate'];
        $aid = $_SESSION['Caraid'];

        $insert = "INSERT INTO Accomodation (aid, gid, sid,
accomodationDate) VALUES ('$aid', '$gid', 'S4', '$order')";
        $B = mysqli_query($db, $insert);

        $insert_ = "INSERT INTO resourceStatus (id, date, isAvailable)
VALUES ('$aid','$order', 0)";
        $A = mysqli_query($db, $insert_ );

        $staff = "INSERT INTO dutySchecule (sid, date, wage) VALUES
('S4', '$order', 1000)";
        mysqli_query($db, $staff);
    }

    header('location: confirmation.php');
}
}

```

Required queries in the assignment pdf

- Monthly guest booking

```

<!-- details for monthly guest booking, given month is entered -->
<?php if(isset($_POST['month_detailsGuest_btn'])) :?>
<?php
$monthNo = mysqli_real_escape_string($db, $_POST['monthGuest']);
$_SESSION['monthNo'] = $monthNo;
if(empty($monthNo) || (!($monthNo >=1 && $monthNo <= 12))) {
    array_push($errors, "Please enter valid month number, ie, between 1
and 12" ); }
else {
    $query=mysqli_query($db, "select * from payment where MONTH(date) =
'$monthNo' " ); }
    if(!empty ($query) && mysqli_num_rows($query) == 0) {
        array_push($errors, "No Booking in this month" );
    } ?>
<?php if(count($errors) == 0) :?>
<table class="styled-table">
    <tr>
        <th colspan="6">
            <h2>Month Payment Details by Guest</h2>
        </th>
    </tr>
    <thead>
    </thead>
    <thead>
        <tr>
            <th>Guest ID</th>
            <th>Date</th>
            <th>Total Bill</th>
        </tr>
    </thead>
    <tbody>
        <?php while(($rows=mysqli_fetch_array($query))) { ?>
        <tr>
            <td><?php echo $rows['GID']; ?></td>
            <td><?php echo $rows['date']; ?></td>
            <td><?php echo $rows['totalBill']; ?></td>
        </tr>
        <?php } ?>
    </tbody>
</table>

```

```

<?php else : ?>
<div class="form">
    <?php include('errors.php') ?>
</div>
<?php endif?>
<?php endif ?>

```

- Monthly room booking

```

<!-- details for monthly room booking, given month is entered -->
<?php if(isset($_POST['month_detailsRoom_btn'])) :?>
<?php
$monthNo = mysqli_real_escape_string($db, $_POST['monthRoom']);
$_SESSION['monthNo'] = $monthNo;

if(empty($monthNo) || (!($monthNo >=1 && $monthNo <= 12))) {
    array_push($errors, "Please enter valid month number, ie, between 1
and 12" ); }
else {
    $query=mysqli_query($db, "select * from Room where MONTH(date) =
'$monthNo' " ); }
    if(!empty ($query) && mysqli_num_rows($query) == 0) {
        array_push($errors, "No Booking in this month" );
    } ?>
<?php if(count($errors) == 0) :?>
<table class="styled-table">
    <tr>
        <th colspan="6">
            <h2>Month Details by Room Booking</h2>
        </th>
    </tr>
    <thead>
    </thead>
    <thead>
        <tr>
            <th>Guest ID</th>
            <th>Room ID</th>
            <th>Date</th>
            <th>Total Bill</th>
        </tr>
    </thead>

```

```

<tbody>
  <?php while(($rows=mysqli_fetch_array($query))) { ?>
    <tr>
      <td><?php echo $rows['GID']; ?></td>
      <td><?php echo $rows['RID']; ?></td>
      <td><?php echo $rows['date']; ?></td>
      <td><?php echo $rows['bill']; ?></td>
    </tr>
  <?php } ?>
</tbody>
</table>
<?php else : ?>
<div class="form">
  <?php include('errors.php') ?>
</div>
<?php endif?>
<?php endif ?>

```

- Monthly food booking

```

<!-- details for monthly food booking, given month is entered -->
<?php if(isset($_POST['month_detailsFood_btn'])) :?>
<?php
$monthNo = mysqli_real_escape_string($db, $_POST['monthFood']);
$_SESSION['monthNo'] = $monthNo;
if(empty($monthNo) || (!($monthNo >=1 && $monthNo <= 12))) {
    array_push($errors, "Please enter valid month number, ie, between 1
and 12" ); }
else {
    $query=mysqli_query($db, "select * from restaurant where MONTH(date) =
'$monthNo'"); }
    if(!empty ($query) && mysqli_num_rows($query) == 0) {
        array_push($errors, "No Booking in this month" );
    } ?>
<?php if(count($errors) == 0) :?>
<table class="styled-table">
  <tr>
    <th colspan="6">
      <h2>Month Details by Food Booking</h2>
    </th>
  </tr>

```

```

<thead>
</thead>
<thead>
  <tr>
    <th>Guest ID</th>
    <th>Date</th>
    <th>Total Bill</th>
  </tr>
</thead>
<tbody>
  <?php while(($rows=mysqli_fetch_array($query))) { ?>
    <tr>
      <td><?php echo $rows['gid']; ?></td>
      <td><?php echo $rows['date']; ?></td>
      <td><?php echo $rows['bill']; ?></td>
    </tr>
    <?php } ?>
  </tbody>
</table>
<?php else : ?>
<div class="form">
  <?php include('errors.php') ?>
</div>
<?php endif?>
<?php endif ?>

```

- Monthly car booking

```

<!-- details for monthly car booking, given month is entered -->
<?php if(isset($_POST['month_details_btnCar'])) :?>
<?php
$monthNo = mysqli_real_escape_string($db, $_POST['monthCar']);
$_SESSION['monthNo'] = $monthNo;
if(empty($monthNo) || (!($monthNo >=1 && $monthNo <= 12))) {
  array_push($errors, "Please enter valid month number, ie, between 1
and 12" ); }
else {
  $query=mysqli_query($db, "select * from Accomodation where
MONTH(accomodationDate) = '$monthNo' " ); }
  if(!empty ($query) && mysqli_num_rows($query) == 0) {
    array_push($errors, "No Booking in this month" );

```

```

    } ?>
<?php if(count($errors) == 0) :?>
<table class="styled-table">
    <tr>
        <th colspan="6">
            <h2>Month Payment Details of Accomodation Booking</h2>
        </th>
    </tr>
    <thead>
    </thead>
    <thead>
        <tr>
            <th>Guest ID</th>
            <th>Staff ID</th>
            <th>Car ID</th>
            <th>Date</th>
            <th>Bill</th>
        </tr>
    </thead>
    <tbody>
        <?php while(($rows=mysqli_fetch_array($query))) { ?>
            <tr>
                <td><?php echo $rows['GID']; ?></td>
                <td><?php echo $rows['SID']; ?></td>
                <td><?php echo $rows['AID']; ?></td>
                <td><?php echo $rows['accomodationDate']; ?></td>
                <td> <?php
                    if($rows['AID'] == 'A1' ) {
                        echo 2000;
                    }
                    else if($rows['AID'] == 'A2') {
                        echo 3000;
                    }
                    else if($rows['AID'] == 'A3') {
                        echo 3000;
                    }
                    else if($rows['AID'] == 'A4') {
                        echo 1500;
                    }
                    else if($rows['AID'] == 'A6') {
                        echo 1000;

```



```

    }
    else echo 1500;
?> </td>
</tr>
<?php } ?>
</tbody>
</table>
<?php else : ?>
<div class="form">
    <?php include('errors.php') ?>
</div>
<?php endif?>
<?php endif ?>

```

- Monthly staff work

```

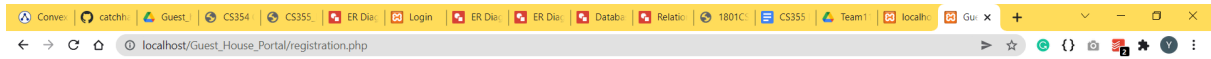
<!-- details for monthly staff work, given month is entered -->
<?php if(isset($_POST['month_detailsStaff_btn'])) :?>
<?php
$monthNo = mysqli_real_escape_string($db, $_POST['monthStaff']);
$_SESSION['monthNo'] = $monthNo;

if(empty($monthNo) || (!($monthNo >=1 && $monthNo <= 12))) {
    array_push($errors, "Please enter valid month number, ie, between 1
and 12" ); }
else {
    $query=mysqli_query($db, "select * from dutySchechule where MONTH(date)
= '$monthNo' " ); }
    if(!empty ($query) && mysqli_num_rows($query) == 0) {
        array_push($errors, "No Booking in this month" );
    } ?>
<?php if(count($errors) == 0) :?>
<table class="styled-table">
    <tr>
        <th colspan="6">
            <h2>Month Payment Details Of Staff</h2>
        </th>
    </tr>
</thead>
</thead>
<thead>

```

```
<tr>
  <th>Staff ID</th>
  <th>Date</th>
  <th>Day Wage</th>
</tr>
</thead>
<tbody>
  <?php while(($rows=mysqli_fetch_array($query))) { ?>
    <tr>
      <td><?php echo $rows['sid']; ?></td>
      <td><?php echo $rows['date']; ?></td>
      <td><?php echo $rows['wage']; ?></td>
    </tr>
    <?php } ?>
  </tbody>
</table>
<?php else : ?>
<div class="form">
  <?php include('errors.php') ?>
</div>
<?php endif?>
<?php endif ?>
```

Screenshots



Guest Registration

Guest ID

Name

Phone Number

Age

Password

Confirm Password

[Register](#)

[Guest Login](#)

[Staff Login](#)

[Admin Login](#)



Guest Login

Guest ID

Password

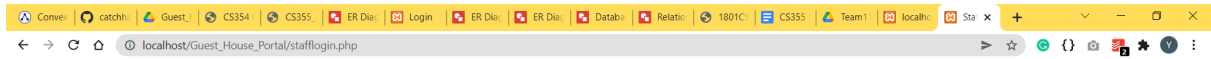
[Login](#)

[Guest Registration](#)

[Staff Login](#)

[Admin Login](#)





Staff Login

Staff ID

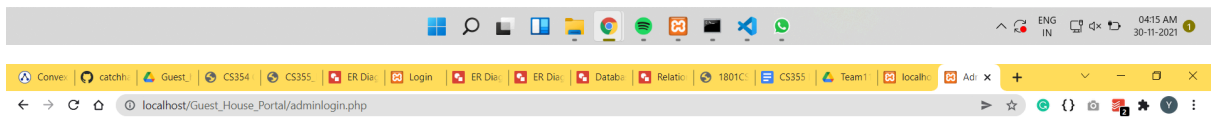
Password

Login

[Guest Registration](#)

[Guest Login](#)

[Admin Login](#)



Admin Login

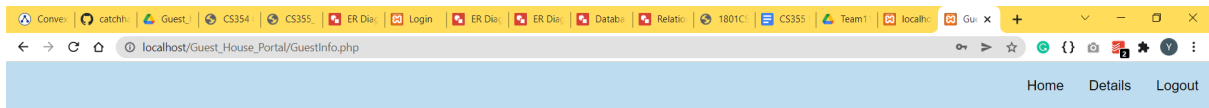
Password

Login

[Guest Registration](#)

[Guest Login](#)

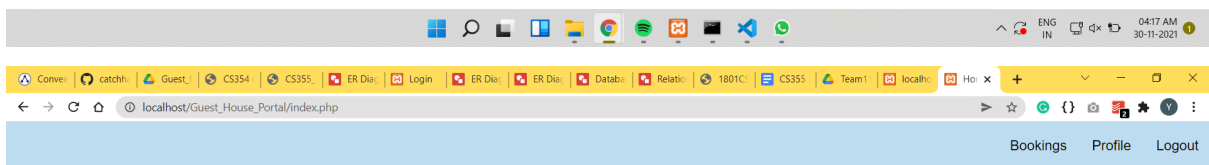
[Staff Login](#)



Guest Information

You are now Logged In

- Guest ID: 1901CS99
- Name: John Doe
- Mobile Number: 12345
- Age: 25



Home Page

Book Rooms
At affordable prices!

Book

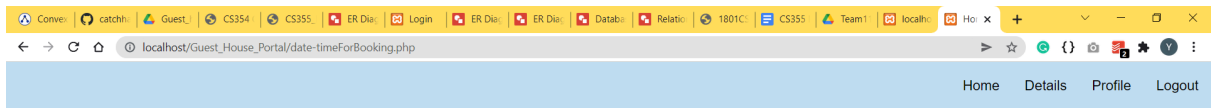
Menu
Hungry?

Order

Book a Cab
Easy Ride!

Book





Select Date

Check-in Date

01-12-2021

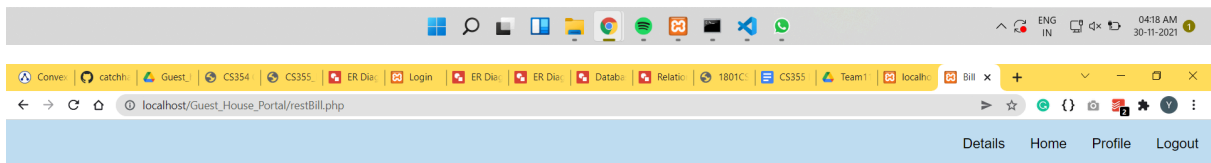


Check-out Date

02-12-2021



Proceed



Bill

- Item: Double Room
- Price: 1500
- Number of days: 1
- Total: ₹ 1500

Place Order



localhost/Guest_House_Portal/payBill.php

DetailsHomeProfileLogout

Payment Details

Full Name

John Doe

Phone Number

12345

Guest ID

1901CS99

Bill

₹ 1500

Name on Card

John Doe

Credit card number

1234-5678-9101

Expiry Month

February

Expiry Year

localhost/Guest_House_Portal/confirmation.php

HomeProfileLogout

Confirmation Page

Payment Confirmed!

[Back](#)

Restaurant Bill

Item	Price
Coffee	0
Extra Cold Coffee	0
Thandai	0
Pancake	0
Pasta	0
Cheese Sandwich	120
Egg Sandwich	0
Daal Makhni	400
Vegetarian Tandoori Platter	0
Chicken Korma	0
Butter Chicken	0
Butter Naan	0
Rumali Roti	0
Total	₹ 520

Place Order

Staff Information

You are now logged in

- Name: Adarsh Shrivastava
- Age: 23
- Mobile Number:34313
- Staff ID: S2
- Password: 1
- Department: Restaurnt
- Role: Restaurant General Manager

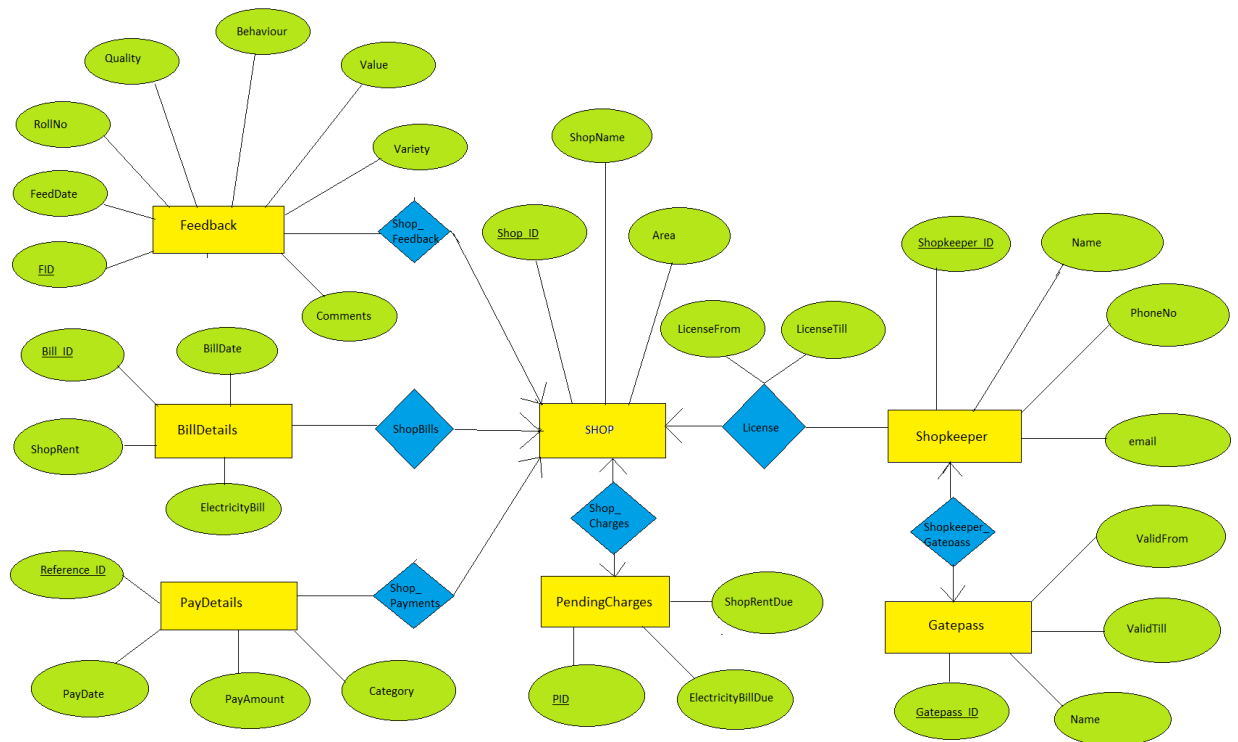
Wage Details

Following are the details of your duty:

Record	
Date	Wage
Total	

Task B: Market Shop Related Services

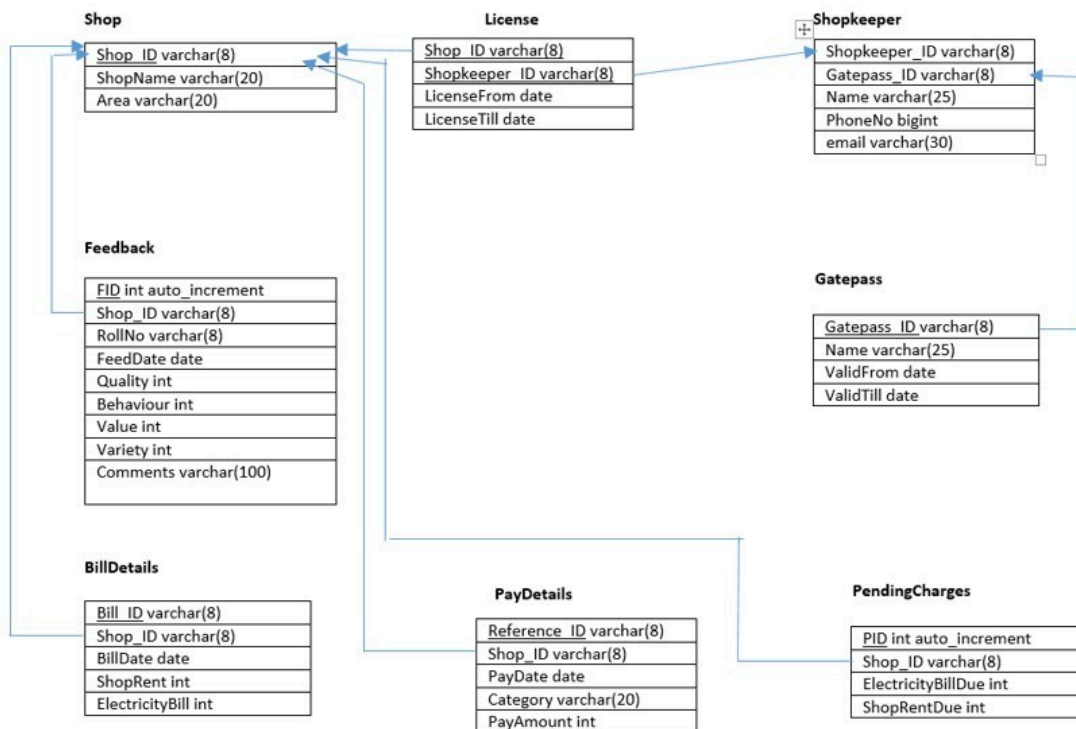
ER Diagram:



- **Shop** relation contains information about different shops on the campus.
- **Shopkeeper** relation contains details about the Shopkeepers of different Shops.
- **Gatepass** relation contains information about gatepass validity of campus visitors.
- **Feedback** relation contains feedback of different shops from students on the basis of parameters like Product/Service quality, Shopkeeper behaviour, Value for money, Variety available. These parameters are rated from 1 to 10 by the students who are identified by unique Roll Numbers.
- **BillDetails** relation contains monthly bills (Electricity Bill and Shop Rent) for each shop.
- **PayDetails** relation contains details of payments from each Shop.
- **PendingCharges** relation contains the pending charges (Electricity Bill and Shop Rent) from each Shop.
- **License** relationship links shopkeepers to their shops along with their license period.

- One shop can have many feedback(from different students), one shop can have many bill details(bills for different months), one shop can have many pay details, one shop can have many shopkeepers, one shopkeeper has one gatepass and one shop has one record for pending charges.

Relational Schema:



- Primary keys are underlined.
- Shop_ID, Shopkeeper_ID, Gatepass_ID , (Shop_ID, Shopkeeper_ID), FID, PID, Bill_ID, Reference_ID uniquely identifies each record in their respective tables hence are defined primary keys.
- Foreign keys are indicated by arrows.
- Shop_ID can be a part of Feedback, License, BillDetails, PayDetails and PendingCharges if and only if it is present in Shop relation hence the foreign keys are defined. Also, the relation License links Shopkeepers to their shops so the Shopkeeper_IDs in License relation must be present in Shopkeeper relation, hence the foreign key is defined.

Creating tables:

```
mysql> create table Shop(  
-> Shop_ID varchar(8) primary key,  
-> ShopName varchar(20),  
-> Area varchar(20));  
  
mysql> create table Gatepass(  
-> Gatepass_ID varchar(8) primary key,  
-> Name varchar(25),  
-> ValidFrom date,  
-> ValidTill date);  
  
mysql> create table Shopkeeper(  
-> Shopkeeper_ID varchar(8) primary key,  
-> Gatepass_ID varchar(8),  
-> Name varchar(25),  
-> PhoneNo bigint,  
-> email varchar(30),  
-> constraint Shopkeeper_fk1 FOREIGN KEY(Gatepass_ID) REFERENCES Gatepass(Gatepass_ID));  
  
mysql> create table License(  
-> Shop_ID varchar(8),  
-> Shopkeeper_ID varchar(8),  
-> LicenseFrom date,  
-> LicenseTill date,  
-> constraint License_pk PRIMARY KEY(Shop_ID, Shopkeeper_ID),  
-> constraint License_fk1 FOREIGN KEY(Shop_ID) REFERENCES Shop(Shop_ID),  
-> constraint License_fk2 FOREIGN KEY(Shopkeeper_ID) REFERENCES Shopkeeper(Shopkeeper_ID));  
  
mysql> create table BillDetails(  
-> Bill_ID varchar(8) primary key,  
-> Shop_ID varchar(8),  
-> BillDate date,  
-> ShopRent int,  
-> ElectricityBill int,  
-> constraint Bill_fk1 foreign key(Shop_ID) references Shop(Shop_ID);
```

```
mysql> create table PayDetails(  
-> Reference_ID varchar(8) primary key,  
-> Shop_ID varchar(8),  
-> PayDate date,  
-> Category varchar(20),  
-> PayAmount int,  
-> constraint pay_fk1 foreign key(Shop_ID) references Shop(Shop_ID));  
  
mysql> create table PendingCharges(  
-> PID int auto_increment primary key,  
-> Shop_ID varchar(8),  
-> ElectricityBillDue int,  
-> ShopRentDue int,  
-> constraint Pending_fk1 foreign key(Shop_ID) references Shop(Shop_ID));  
  
mysql> create table Feedback(  
-> FID int auto_increment primary key,  
-> Shop_ID varchar(8),  
-> RollNo varchar(8),  
-> FeedDate date,  
-> Quality int,  
-> Behaviour int,  
-> Value int,  
-> Variety int,  
-> Comments varchar(100));  
  
//Two additional tables for login and feedback purpose  
  
mysql> create table Supervisor(  
-> Supervisor_ID varchar(8) primary key,  
-> passwd varchar(255));  
  
mysql> create table studpasswd(  
-> RollNo varchar(8) primary key,  
-> passwd varchar(255));
```

Adding Indices:

```
mysql> alter table BillDetails  
-> add index idx_bill(Shop_ID);  
  
mysql> alter table PendingCharges  
-> add index idx_pending(Shop_ID);  
  
mysql> alter table Feedback  
-> add index idx_feed(Shop_ID);
```

Procedures:

1) To update Electricity bill payment details in the database:

(Although bills are paid by shopkeepers of different shops but the details can be updated only by the supervisor when he logs in)

```
mysql> delimiter $$
mysql> create procedure PayElectricityBill(in _ref_id varchar(8),in _shop_id varchar(8),in _paydate date,in _amt int)
-> begin
-> insert into PayDetails values(_ref_id,_shop_id,_paydate,'Electricity',_amt);
-> update PendingCharges set ElectricityBillDue = ElectricityBillDue - _amt where Shop_ID=_shop_id;
-> end
-> $$
Query OK, 0 rows affected (0.03 sec)
```

2) To update Shop Rent payment details in the database:

```
mysql> create procedure PayShopRent(in _ref_id varchar(8),in _shop_id varchar(8),in _paydate date,in _amt int)
-> begin
-> insert into PayDetails values(_ref_id,_shop_id,_paydate,'ShopRent',_amt);
-> update PendingCharges set ShopRentDue = ShopRentDue - _amt where Shop_ID=_shop_id;
-> end
-> $$
Query OK, 0 rows affected (0.05 sec)
```

Triggers:

Whenever a bill (for a month) is inserted in BillDetails relation, we update the PendingCharges relation by adding the bill amounts in ElectricityBillDue and ShopRentDue attributes for the particular month. Also if a shop has pending charges, a penalty of Rs 50 is added.

```
mysql> create trigger before_billdetails_insert
-> before insert
-> on billdetails for each row
-> begin
-> declare rowcount int;
-> declare penalty,ebd,srd int;
-> select count(*) into rowcount from pendingcharges
-> where Shop_ID = new.Shop_ID;
-> if(rowcount=0) then
-> insert into pendingcharges(Shop_ID,ElectricityBillDue,ShopRentDue) values
-> (new.Shop_ID,new.ElectricityBill,new.ShopRent);
-> else
-> select ElectricityBillDue into ebd from pendingcharges
-> where Shop_ID = new.Shop_ID;
-> select ShopRentDue into srd from pendingcharges
-> where Shop_ID = new.Shop_ID;
-> set penalty = 50;
-> if(ebd > 0) then
-> set ebd = ebd + new.ElectricityBill + penalty;
-> else
-> set ebd=ebd + new.ElectricityBill;
-> end if;
-> if(srd > 0) then
-> set srd = srd + new.ShopRent + penalty;
-> else
-> set srd = srd + new.ShopRent;
-> end if;
-> update pendingcharges set ElectricityBillDue = ebd where Shop_ID = new.Shop_ID;
-> update pendingcharges set ShopRentDue = srd where Shop_ID = new.Shop_ID;
-> end if;
-> end
-> $$
```

Queries:

1) Shop Details of different areas of the campus:

```
$query="select * from Shop";
$result=mysqli_query($db,$query);

echo "
<table border=\"5\" cellpadding=\"5\" bordercolor=\"#808080\" bgcolor=\"#C0C0C0\">
<tr>
<td>Shop ID</td>
<td>Shop Name</td>
<td>Area</td>
</tr>";

while($user=mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>" . $user['Shop_ID'] . "</td>";
    echo "<td>" . $user['ShopName'] . "</td>";
    echo "<td>" . $user['Area'] . "</td>";
    echo "<tr>";
}
echo "</table>";
```

2) Details of shop keepers and their security pass validity:

Query for Supervisors:

```
$query="select * from shopkeeper natural join gatepass";
$result=mysqli_query($db,$query);

echo "





```

Query for Shopkeepers:

```
$query="select * from shopkeeper natural join gatepass where Shopkeeper_ID='$Shopkeeper_ID'";
$result=mysqli_query($db,$query);
$user=mysqli_fetch_assoc($result);

echo "Name: " . $user['Name'] . "<br>";
echo "Phone No.: " . $user['PhoneNo'] . "<br>";
echo "Email: " . $user['email'] . "<br>";
echo "Gatepass ID: " . $user['Gatepass_ID'] . "<br>";
echo "Valid From: " . $user['ValidFrom'] . "<br>";
echo "Valid Till: " . $user['ValidTill'] . "<br>";

$curdate=date("Y-m-d");
$nextmonth=date("Y-m-d", strtotime($curdate. ' + 30 days'));
if($user['ValidTill']<$curdate)
echo "<div style=\"color:red;\>Your Gatepass Validity has expired</div>" . "<br>";
else if($user['ValidTill']<$nextmonth)
echo "<div style=\"color:red;\>Your Gatepass Validity will expire within 30 days</div>" . "<br>";
```

3) Details of shop licenses:

For Supervisors:

```
$query="select * from license";
$result=mysqli_query($db,$query);

echo "





```

For Shopkeepers:

```
echo "<br><h4>Shop License Details</h4><br>";
$query="select * from license where Shopkeeper_ID='$Shopkeeper_ID'";
$result=mysqli_query($db,$query);
$user=mysqli_fetch_assoc($result);

echo "Shop ID: " . $user['Shop_ID'] . "<br>";
echo "License From: " . $user['LicenseFrom'] . "<br>";
echo "License Till " . $user['LicenseTill'] . "<br>";

$curdate=date("Y-m-d");
if($user['LicenseTill']<$curdate)
echo "Your Shop License has expired" . "<br>";
```

4) View Bill Details:

For Supervisors:

```
$query="select * from BillDetails";
$result=mysqli_query($db,$query);

echo "






while($user=mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>" . $user['Bill_ID'] . "</td>";
    echo "<td>" . $user['Shop_ID'] . "</td>";
    echo "<td>" . $user['BillDate'] . "</td>";
    echo "<td>" . $user['ShopRent'] . "</td>";
    echo "<td>" . $user['ElectricityBill'] . "</td>";
    echo "<tr>";
}
echo "</table>";
```

For Shopkeepers:

```
$query="select * from BillDetails natural join License where Shopkeeper_ID='$Shopkeeper_ID'";
$result=mysqli_query($db,$query);

echo "






while($user=mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>" . $user['Bill_ID'] . "</td>";
    echo "<td>" . $user['BillDate'] . "</td>";
    echo "<td>" . $user['ShopRent'] . "</td>";
    echo "<td>" . $user['ElectricityBill'] . "</td>";
    echo "<tr>";
}
echo "</table>";
```

5) View Payment Details:

For Supervisors:

```
$query="select * from PayDetails";
$result=mysqli_query($db,$query);

echo "
<table border=\"5\" cellpadding=\"5\" bordercolor=\"#808080\" bgcolor=\"#C0C0C0\">
<tr>
<td>Reference ID</td>
<td>Shop ID</td>
<td>Category</td>
<td>Payment Date</td>
<td>Amount Paid</td>
</tr>";

while($user=mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>" . $user['Reference_ID'] . "</td>";
    echo "<td>" . $user['Shop_ID'] . "</td>";
    echo "<td>" . $user['Category'] . "</td>";
    echo "<td>" . $user['PayDate'] . "</td>";
    echo "<td>" . $user['PayAmount'] . "</td>";
    echo "<tr>";
}
echo "</table>";
```

For Shopkeeper:

```
echo "<br><h4>Previous Payment Details</h4><br>";
$query="select * from PayDetails natural join License where Shopkeeper_ID='$Shopkeeper_ID'";
$result=mysqli_query($db,$query);

echo "
<table border=\"5\" cellpadding=\"5\" bordercolor=\"#808080\" bgcolor=\"#C0C0C0\">
<tr>
<td>Reference ID</td>
<td>Category</td>
<td>Payment Date</td>
<td>Amount Paid</td>
</tr>";

while($user=mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>" . $user['Reference_ID'] . "</td>";
    echo "<td>" . $user['Category'] . "</td>";
    echo "<td>" . $user['PayDate'] . "</td>";
    echo "<td>" . $user['PayAmount'] . "</td>";
    echo "<tr>";
}
echo "</table>";
```

5) View Pending Charges:

For Supervisors:

```
$query="select * from PendingCharges";
$result=mysqli_query($db,$query);

echo "





```

For Shopkeeper:

```
$query="select * from PendingCharges natural join license where Shopkeeper_ID='$Shopkeeper_ID'";
$result=mysqli_query($db,$query);
$user=mysqli_fetch_assoc($result);

if(mysqli_num_rows($result) > 0)
{
    echo "Shop ID: " . $user['Shop_ID'] . "<br>";
    echo "Electricity Bill Due: " . $user['ElectricityBillDue'] . "<br>";
    echo "Shop Rent Due" . $user['ShopRentDue'] . "<br>";
}
```

6) Performance Summary through Feedback:

```
$query="select Shop_ID,avg(Quality),avg(Behaviour),avg(Value),avg(Variety),(avg(Quality)+avg(Behaviour)+avg(Value)+avg(Variety))/4 as
OverallAverage from Feedback group by Shop_ID";
$result=mysqli_query($db,$query);

echo "





```

7) Adding Bills and updating Bill payments:

```
if(isset($_POST['AddBill']))
{
    $Shop_ID=mysqli_real_escape_string($db,$_POST['Shop_ID']);
    $Bill_ID=mysqli_real_escape_string($db,$_POST['Bill_ID']);
    $BillDate=htmlentities($_POST['BillDate']);
    $BillDate = date('Y-m-d', strtotime($BillDate));
    $ShopRent=$_POST['ShopRent'];
    $ElectricityBill=$_POST['ElectricityBill'];

    $query="select * from Shop where Shop_ID='$Shop_ID'";
    $result=mysqli_query($db,$query) or die(mysql_error());
    $user=mysqli_fetch_assoc($result);

    if(mysqli_num_rows($result) > 0)
    {
        $query="insert into BillDetails values('$Bill_ID','$Shop_ID','$BillDate',$ShopRent,$ElectricityBill)" or die(mysql_error());
        $result=mysqli_query($db,$query) or die(mysql_error());
        echo "Bill details added successfully";
    }
    else
    {
        echo "Shop ID does not exist<br>";
    }
}
```

```
if(isset($_POST['PayElectricityBill']))
{
    $Shop_ID=mysqli_real_escape_string($db,$_POST['Shop_ID']);
    $Reference_ID=mysqli_real_escape_string($db,$_POST['Reference_ID']);
    $PayDate=htmlentities($_POST['PayDate']);
    $PayDate = date('Y-m-d', strtotime($PayDate));
    $PayAmount=$_POST['PayAmount'];

    $query="select * from Shop where Shop_ID='$Shop_ID'";
    $result=mysqli_query($db,$query) or die(mysql_error());

    if(mysqli_num_rows($result) > 0)
    {
        $query="call PayElectricityBill('$Reference_ID','$Shop_ID','$PayDate',$PayAmount)";
        $result=mysqli_query($db,$query) or die(mysql_error());
        echo "Electricity Bill Payment details updated successfully";
    }
    else
    {
        echo "Shop ID does not exist<br>";
    }
}
```

```
if(isset($_POST['PayShopRent']))
{
    $Shop_ID=mysqli_real_escape_string($db,$_POST['Shop_ID']);
    $Reference_ID=mysqli_real_escape_string($db,$_POST['Reference_ID']);
    $PayDate=htmlentities($_POST['PayDate']);
    $PayDate = date('Y-m-d', strtotime($PayDate));
    $PayAmount=$_POST['PayAmount'];
    $query="select * from Shop where Shop_ID='$Shop_ID'";
    $result=mysqli_query($db,$query) or die(mysql_error());

    if(mysqli_num_rows($result) > 0)
    {
        $query="call PayShopRent('$Reference_ID','$Shop_ID','$PayDate',$PayAmount)";
        $result=mysqli_query($db,$query) or die(mysql_error());
        echo "Shop Rent Payment details updated successfully";
    }
    else
    {
        echo "Shop ID does not exist<br>";
    }
}
```

Web Interface Screenshots:

Home Page:

Market Shop Related Services

[Shopkeeper Login](#)

[Supervisor Login](#)

[Submit Shop Feedback](#)



Shopkeeper Login Page:

Market Shop Related Services

Shopkeeper Login

Shopkeeper ID

Password

[Go to Home Page](#)

Shopkeeper Welcome Page

Market Shop Related Services

Welcome K05

Gatepass Validity

Name: Raj
Phone No.: 9834567480
Email: raj05@gmail.com
Gatepass ID: G15
Valid From: 2021-09-06
Valid Till: 2022-09-06

Shop License Details

Shop ID: S02
License From: 2020-06-21
License Till 2022-06-21

Bill Details

Bill ID	Bill Date	Shop Rent	Electricity Bill
B002	2021-11-10	600	850

Previous Payment Details

Reference ID	Category	Payment Date	Amount Paid
RF01	Electricity	2021-11-18	500

Pending Charges

Shop ID: S02
Electricity Bill Due: 350
Shop Rent Due600
[Logout](#)

Supervisor Login Page:

Market Shop Related Services

Supervisor Login

Supervisor ID
Password

[Go to Home Page](#)

Supervisor Welcome Page:

Market Shop Related Services

Welcome SUP01

[Add Bill/Update Bill Payment Details](#)

Shop Details

Shop ID	Shop Name	Area
S01	Nescafe	Old Boys Hostel
S02	Nescafe	Academic Area
S03	Night Canteen	Old Boys Hostel
S04	Amul	Food Court
S05	Krishna Stationary	Food Court
S06	Sriram Fastfood	Food Court
S07	Cycle Shop	Tennis Court
S08	Barber Shop	Tennis Court
S09	Vegetable Shop	Gate-2
S10	Kirana Store	Gate-2

Details of Shopkeepers and their Gatepass Validity

Shopkeeper ID	Name	PhoneNo	email	Gatepass ID	Valid From	Valid Till
K01	Arpit	9334567480	arpit01@gmail.com	G11	2021-09-06	2022-09-06
K02	Mohan	9334597480	mohan02@gmail.com	G12	2020-08-06	2021-08-06
K03	Harsh	9334907480	harsh03@gmail.com	G13	2021-07-05	2022-07-05
K04	Shivam	8734554480	shivam04@gmail.com	G14	2020-11-12	2021-11-12
K05	Raj	9834567480	raj05@gmail.com	G15	2021-09-06	2022-09-06
K06	Hemant	7634567480	hemant06@gmail.com	G16	2021-05-08	2022-05-08
K07	Shashwat	7934567480	shashwat07@gmail.com	G17	2020-12-06	2021-12-06
K08	Yash	8234567480	yash08@gmail.com	G18	2019-11-12	2020-11-11
K09	Vivek	9934567480	vivek09@gmail.com	G19	2021-04-06	2022-04-06
K10	Vishal	9434567480	vishal10@gmail.com	G20	2021-09-15	2022-09-15
K11	Mohit	7734567480	mohit11@gmail.com	G01	2021-09-06	2022-09-06
K12	Rohit	8336667480	rohit12@gmail.com	G02	2020-08-06	2021-08-06
K13	Shobhit	9234567480	shobhit13@gmail.com	G03	2021-07-05	2022-07-05
K14	Ritwik	9134567480	ritwik14@gmail.com	G04	2020-11-12	2021-11-12
K15	Yash	7834568980	yash15@gmail.com	G05	2021-09-06	2022-09-06

Shop License Details

Shop ID	Shopkeeper ID	License From	License Till
S01	K03	2020-06-22	2022-06-22
S01	K04	2020-06-18	2022-06-18
S02	K05	2020-06-21	2022-06-21
S03	K06	2020-06-17	2022-06-17
S04	K07	2019-10-05	2021-10-05
S04	K08	2020-06-21	2022-06-21
S04	K09	2020-07-05	2022-07-05
S05	K10	2019-12-05	2021-12-05
S06	K11	2020-06-26	2022-06-26
S07	K12	2020-06-15	2022-06-15
S08	K13	2020-06-22	2022-06-22
S09	K14	2020-07-10	2022-07-10
S09	K15	2020-07-02	2022-07-02
S10	K01	2020-07-02	2022-07-02
S10	K02	2020-07-12	2022-07-12

Bill Details

Bill ID	Shop ID	Bill Date	Shop Rent	Electricity Bill
B001	S01	2021-11-15	500	750
B002	S02	2021-11-10	600	850
B003	S01	2021-12-15	500	700
B004	S03	2021-10-20	550	900
B005	S04	2021-11-10	1000	1500
B006	S05	2021-11-16	1250	1000
B007	S06	2021-11-21	1500	1250
B008	S07	2021-11-19	1750	1500
B009	S08	2021-11-24	850	750
B010	S09	2021-11-26	2500	1500
B011	S10	2021-11-24	700	700

Previous Payment Details

Reference ID	Shop ID	Category	Payment Date	Amount Paid
RF01	S02	Electricity	2021-11-18	500
RF02	S03	Electricity	2021-10-30	900
RF03	S03	ShopRent	2021-10-31	550
RF04	S04	Electricity	2021-11-25	1000
RF05	S04	ShopRent	2021-11-25	1000
RF06	S06	Electricity	2021-11-30	600
RF07	S07	Electricity	2021-11-30	500
RF08	S08	ShopRent	2021-11-30	450
RF09	S09	Electricity	2021-11-29	400
RF10	S09	ShopRent	2021-11-29	550

Pending Charges

Shop ID	Electricity Bill Due	Shop Rent Due
S01	1500	1050
S02	350	600
S03	0	0
S04	500	0
S05	1000	1250
S06	650	1500
S07	1000	1750
S08	750	400
S09	1100	1950
S10	700	700

Performance Summary

Shop ID	avg(Quality)	avg(Behaviour)	avg(Value)	avg(Variety)	OverallAverage
S01	8.3333	6.6667	8.3333	9.3333	8.16666667
S02	10.0000	9.5000	9.5000	10.0000	9.75000000
S03	8.5000	8.5000	9.0000	8.5000	8.62500000
S06	7.5000	8.5000	8.5000	6.5000	7.75000000
S08	6.5000	7.0000	7.5000	7.0000	7.00000000
S10	9.0000	9.0000	9.0000	10.0000	9.25000000

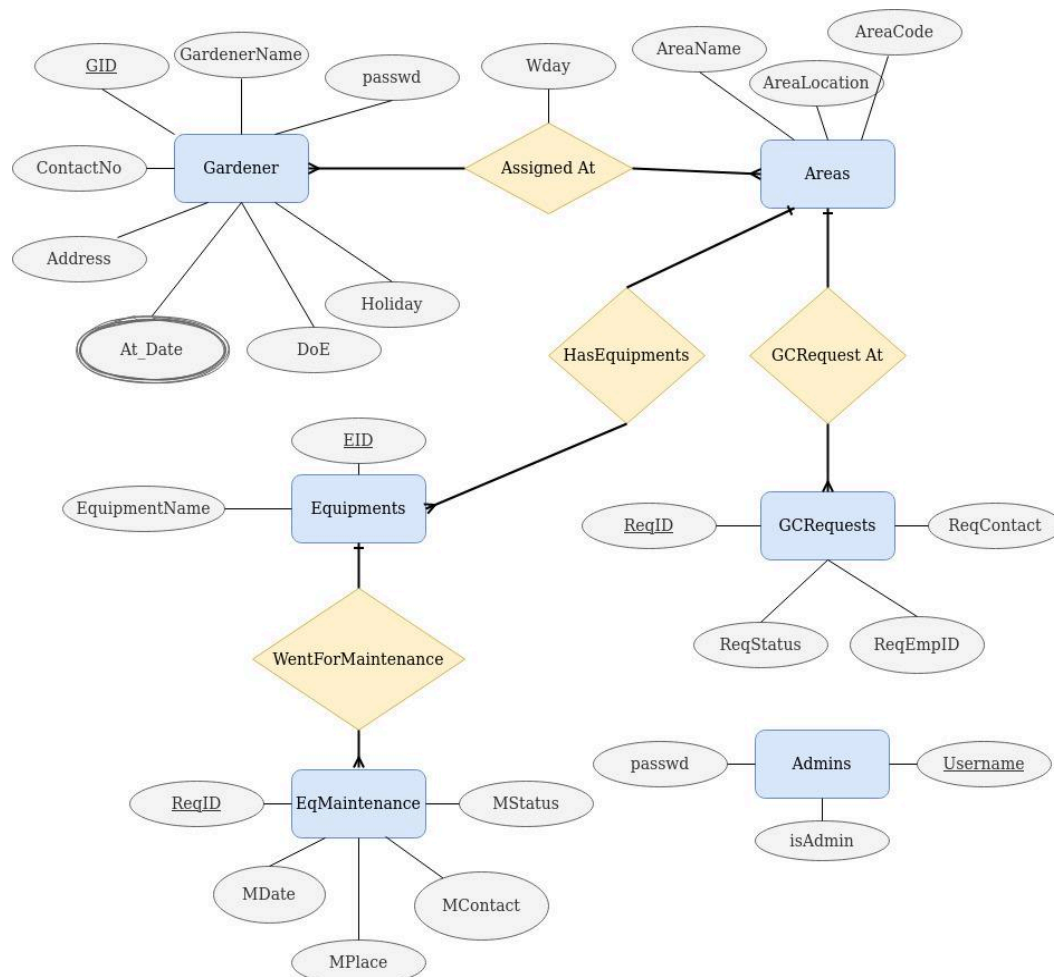
[Logout](#)

TASK C: - Landscaping Related Activities

We were required to create a module in our mini project which was capable of doing the following tasks: -

- Maintaining gardeners details
- Identifying campus areas,
- Gardeners assignments,
- Duty roster(Monthly),
- Equipment stock and maintenance details,
- Handling grass cutting requests

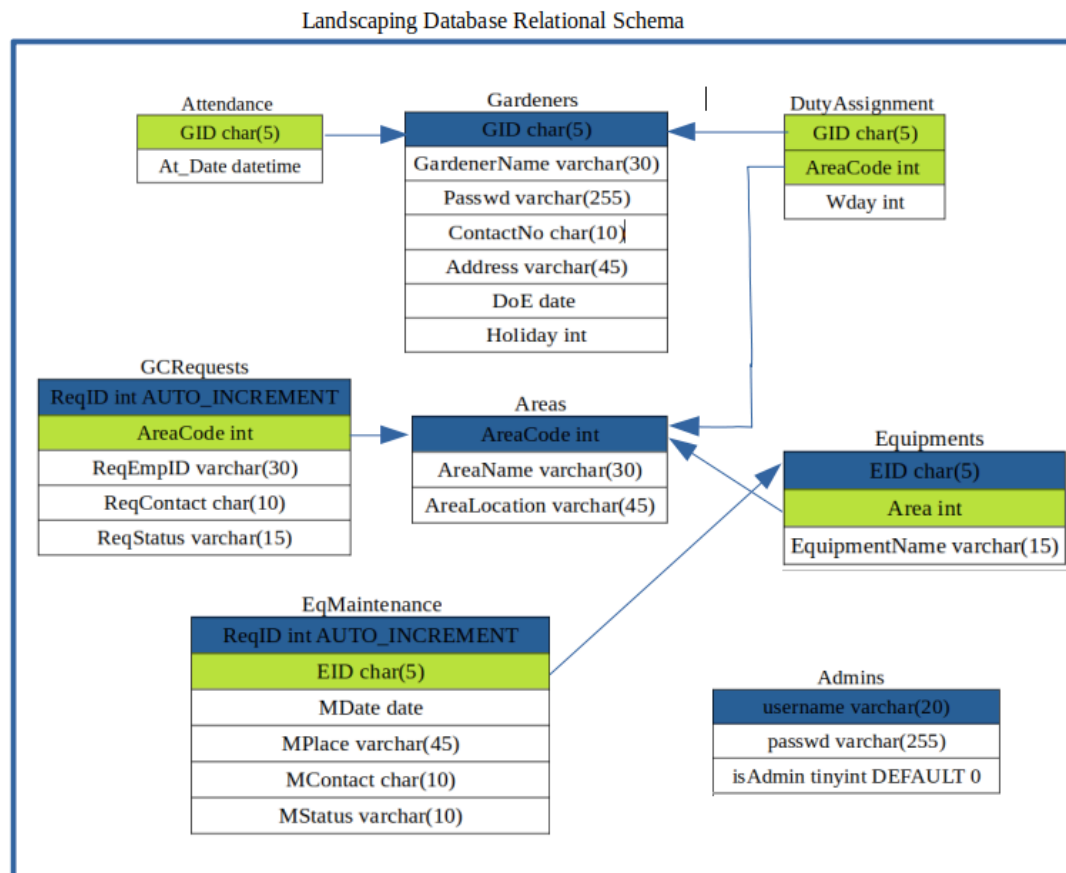
1.) Entity-Relation Model for A database capable of doing above mentioned tasks: -



In the above ER Diagram some special points are as follows: -

- I. Gardeners Entity has 1 multivalued attribute At_Date which stores that specific gardener's attendance. In the relational model, I have stored it in a different table.
- II. Relationship "Assigned At" relates that a given gardener with ID GID is assigned at area with given AreaCode.
 Since multiple gardeners can be assigned to the same area and A gardener can work on multiple areas, it is a Many to Many Relationship.
- III. Relationship "Has Equipment" relates 1 area with all the equipment that given area has. Since each equipment belongs to a single area it is a one to many relationship.
- IV. Relationship "GCRequest At" related Grass Cutting Request to its area it is from, so each request belongs to a single area but A single area can have many requests, so it is many to 1 relationship.
- V. Relationship "Went for Maintenance" Relates equipment to its maintenance log, since each equipment can be sent for maintenance multiple times it is a one to many relationship.
- VI. Admins entity is not related to any other Entity as it just stores login username and password for Administrator accounts(supervisors).

2.) Relational Model : -



Primary Keys for Relationship Tables: -

Attendance: - GID+ At_Date

DutyAssignment: - GID + Wday

For all other tables, Primary Key is in blue colour and foreign Keys are in green.

The Tables are optimised as we can see.

They are in 1NF as all values in all tables are atomic.

They are also optimised to be in 2NF form.

They are in 3NF form as there is no transitive partial dependency for non-prime attributes in the tables

They are also in BCNF as for every attribute it is only dependent on the primary key and Hence all LHS of Functional dependency will be SuperKey.

3) Queries to Create and populate above relational schema in our mysql Server: -
--

```
$ mysql -u root - p
```

QUERIES: -

Create Database and Use it: -

```
> CREATE DATABASE Landscaping;  
> USE Landscaping;  
> GRANT ALL PRIVILEGES ON Landscaping.* TO 'scot'@'localhost';
```

Creating all required tables: -

1. **Gardeners Table** : This table stores all the gardener details like :gardener ID, name, address, contactNo, allotted holiday in week (0-6) ,date of employment etc.
QUERY: -

```
> CREATE TABLE Gardeners (  
  GID char(5) PRIMARY KEY,  
  GardenerName varchar(30) NOT NULL,  
  passwd varchar(255) ,  
  ContactNo char(10) ,  
  Address varchar(45) ,  
  DoE date NOT NULL,  
  Holiday int );
```

2. **Attendance Table** : this stores GID of gardeners paired with dates they were present at.

QUERY: -

```

> CREATE TABLE Attendance (
  GID char(5) NOT NULL,
  At_Date datetime NOT NULL,
  PRIMARY KEY (GID,At_Date),
  CONSTRAINT fk_Attendance_1 FOREIGN KEY (GID) REFERENCES
Gardeners (GID)
> CREATE TABLE DutyAssignment (
  GID char(5) NOT NULL,
  AreaCode int NOT NULL,
  Wday int NOT NULL,
  PRIMARY KEY (GID,Wday),
  KEY fk_DutyAssignment_2_idx (AreaCode),
  CONSTRAINT fk_DutyAssignment_1 FOREIGN KEY (GID) REFERENCES
Gardeners (GID),
  CONSTRAINT fk_DutyAssignment_2 FOREIGN KEY (AreaCode) REFERENCES
Areas (AreaCode)
);

```

3. **DutyAssignment Table:** This relation stores the tuples(Wday (weekday), GID, AreaCode) which represent that on a given weekday , gardener with GID works in Area with this AreaCode.

QUERY: -

```

> CREATE TABLE DutyAssignment (
  GID char(5) NOT NULL,
  AreaCode int NOT NULL,
  Wday int NOT NULL,
  PRIMARY KEY (GID,Wday),
  KEY fk_DutyAssignment_2_idx (AreaCode),
  CONSTRAINT fk_DutyAssignment_1 FOREIGN KEY (GID) REFERENCES Gardeners (GID),
  CONSTRAINT fk_DutyAssignment_2 FOREIGN KEY (AreaCode) REFERENCES Areas
(AreaCode)
);

```

4. **Areas Table :** This table stores details about how campus is divided into different areas.

QUERY: -

```

> CREATE TABLE Areas (
  AreaCode int PRIMARY KEY,
  AreaName varchar(30) NOT NULL,
  AreaLocation varchar(45) NOT NULL,
);

```

5. **GRequests Table:** This table stores the various grass cutting requests submitted by employees or other officials residing on campus.

QUERY: -

```

> CREATE TABLE GRequests (

```

```

ReqID int AUTO_INCREMENT PRIMARY KEY,
AreaCode int NOT NULL,
ReqEmpID varchar(30) NOT NULL,
ReqContact char(10) NOT NULL,
ReqStatus varchar(15),
KEY fk_GCRequests_1_idx (AreaCode),
CONSTRAINT fk_GCRequests_1 FOREIGN KEY (AreaCode) REFERENCES Areas(AreaCode)
);

```

6. **Equipments Relation:** This relation stores all the currently available equipment and tools on campus and their allotted areas.

QUERY: -

```

> CREATE TABLE Equipments (
    EID char(5) PRIMARY KEY,
    Area int NOT NULL,
    EquipmentName varchar(15) NOT NULL,
    KEY fk_Equipments_1_idx (Area),
    CONSTRAINT fk_Equipments_1 FOREIGN KEY (Area) REFERENCES
Areas (AreaCode));

```

7. **EqMaintenance Relation :-** This relation stores maintenance logs for equipment who have gone for repairing or maintenance.

QUERY: -

```

> CREATE TABLE EqMaintenance (
    ReqID int AUTO_INCREMENT PRIMARY KEY,
    EID char(5) NOT NULL,
    MDate date NOT NULL,
    MPlace varchar(45) NOT NULL,
    MContact char(10) NOT NULL,
    MStatus varchar(10) NULL,
    KEY fk_EqMaintenance_1_idx (EID),
    CONSTRAINT fk_EqMaintenance_1 FOREIGN KEY (EID) REFERENCES Equipments (EID));

```

8. **Admins Table:** This simply stores the username and password for admin accounts registered through the portal.

QUERY: -

```

> CREATE TABLE `Admins` (
    username varchar(20) PRIMARY KEY,
    passwd varchar(255) NOT NULL,
    isAdmin tinyint NOT NULL DEFAULT '0',
);

```

isAdmin is made 1 by another admin after verification. Admin accounts are useless unless they are verified and made active by another Administrator. First account is made administrator manually through sql query.

Populating all above relations through CSV files containing sufficient records for testing and running queries.

```
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/gardeners.csv'
INTO TABLE Gardeners FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1
LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/Areas.csv' INTO
TABLE Areas FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE 1 LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/Attendance.csv'
INTO TABLE Attendance FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE
1 LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/DutyAssignment.csv'
INTO TABLE DutyAssignment FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n'
IGNORE 1 LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/Equipments.csv'
INTO TABLE Equipments FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE
1 LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/EqMaintenance.csv'
INTO TABLE EqMaintenance FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n'
IGNORE 1 LINES;
> LOAD DATA LOCAL INFILE '/home/abhay376/Desktop/TermProject/GCRequests.csv'
INTO TABLE GCRequests FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n' IGNORE
1 LINES;
```

OUTPUT:-

QUERY	OUTPUT
SELECT COUNT(*) FROM Gardeners;	15
SELECT COUNT(*) FROM Areas;	15
SELECT COUNT(*) FROM Attendance;	45
SELECT COUNT(*) FROM DutyAssignment;	17
SELECT COUNT(*) FROM Equipments;	16
SELECT COUNT(*) FROM EqMaintenance;	5
SELECT COUNT(*) FROM GCRequests;	5
SELECT COUNT(*) FROM Admins;	2

4) QUERIES (php-mysql) for required tasks used in web interface: -

All following queries have php variables after proper validation and after converting into mysql escape sequence to protect from sql injection attacks.

1) Registration for Admin Account: -

```

$query= "INSERT INTO Admins(username,passwd) VALUES
('$username','$hashed_password')";

if ($result = mysqli_query($db,$query))
{
    echo "<script>alert('Successfully Registered. Please Login!');
window.location.href='./adminLogin.php';</script>";
}

```

2) Login for Admin : -

```

$query="SELECT * FROM Admins WHERE username='$username'";

if ($result = mysqli_query($db,$query))
{
    $nr=mysqli_num_rows($result);
    if($nr==0)
        $err="This User does not exist.Please enter correct
Username.";
    else
    {
        $row=mysqli_fetch_array($result);
        $hashed_password= $row['passwd'];
        if(password_verify($passwd, $hashed_password))
        {
            session_start();
            $_SESSION["username"] = $username;
            $_SESSION["loggedin"] = true;
            $_SESSION["admin"]=$row['isAdmin'];
            //Redirect user to welcome page
            echo "Login Successful!";
            header("location: adminWelcome.php");
        }
        else{
            $err="Incorrect credentials given.";
        }
    }
}
}

```

3) Registration for Gardener Account : -

```

$hashed_password = password_hash($passwd, PASSWORD_DEFAULT);
$query= "UPDATE Gardeners SET passwd ='$hashed_password' WHERE GID='$GID'";

if ($result = mysqli_query($db,$query)){
    echo "<script>alert('Successfully Registered. Please Login!');

```

```
window.location.href='./gardLogin.php';</script>";
}
```

4) Login for Gardener: -

```
$query="SELECT passwd FROM Gardeners WHERE GID='$GID'";
if ($result = mysqli_query($db,$query)) {
    $nr=mysqli_num_rows($result);
    if($nr==0)
        $err="This Gardener does not exist.Please enter correct Gardener ID.";
    else{
        $row=mysqli_fetch_array($result);
        if($row['passwd']== "") {
            $err="This account is not yet registered.";
        }
        else{
            $hashed_password= $row['passwd'];
            if(password_verify($passwd, $hashed_password)){
                session_start();
                $_SESSION["GID"] = $GID;
                $_SESSION["loggedin"] = true;
                //Redirect user to welcome page
                echo "Login Successful!";
                header("location: gardProfile.php");
            }
            else{
                $err="Incorrect credentials given.";
            }
        }
    }
}
```

5) Fetching Gardener Details :-

```
$query = "SELECT * FROM Gardeners WHERE GID ='$GID'";
```

6) Updating Gardener Contact Number: -

```
$query = "SELECT * FROM Gardeners WHERE GID ='$GID'";
```

7) Viewing Gardener's Schedule : -

```
$GID= $_SESSION["GID"];
$dayofweek = array("Sunday", "Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday");
```

```

$query = "SELECT Wday, AreaName FROM DutyAssignment JOIN Areas ON
DutyAssignment.AreaCode=Areas.AreaCode WHERE GID ='$GID'";

if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"><th align="center" >Day</th><th
align="center">Area</th></tr>';

        // output data of each row
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center" ><td align="center">'
.$dayofweek[$row["Wday"]]. '</td><td
align="center">'.$row["AreaName"].'</td></tr>';
        }
        echo "</table>";
        echo "</div>";
    }
    else{
        echo "No Records!";
    }
}

```

8) Posting Grass Cutting Requests: -

```

$query= "INSERT INTO GCRequests (AreaCode, ReqEmpID, ReqContact, ReqStatus)
VALUES ('$AreaCode', '$empID', '$ReqContact', 'Submitted')";

if ($result = mysqli_query($db,$query)){
    $id = mysqli_insert_id($db);
    $success= "Successfully Submitted your request. Your Request ID is :
'$id'";
    echo "<script>alert('$success')</script>";
}

```

9) Viewing Grass Cutting Requests :-

```

$query = "SELECT * FROM GCRequests";
if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> ReqID
</th> <th align="center"> Area Code </th> <th align="center"> ReqEmpID </th> <th
align="center"> ReqContact </th> <th align="center"> ReqStatus </th> </tr>';

        // output data of each row
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center"> <td align="center">' . $row["ReqID"].'</td>

```

```

<td align="center">' . $row["AreaCode"] . '</td> <td
align="center">' . $row["ReqEmpID"] . '</td> <td align="center">'
. $row["ReqContact"] . '</td> <td align="center">' . $row["ReqStatus"] .
'</td></tr>';
    }
    echo "</table>";
    echo "</div>";
}
}

```

10) Updating Req Status after finishing job : -

```

$query= "UPDATE GCRequests SET ReqStatus='$ReqStatus' WHERE ReqID='$ReqID'";

if ($result = mysqli_query($db,$query)){
    header("location: viewGCReq.php");
}

```

11) Listing all Gardeners currently employed : -

```

$query = "SELECT * FROM Gardeners";
$dayofweek = array("Sunday", "Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday");

if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> GID </th>
<th align="center"> GardenerName</th> <th align="center"> ContactNo</th><th
align="center"> Address </th><th align="center"> DoE </th><th align="center">
Holiday </th></tr>';

        // output data of each row
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center"> <th align="center"> ' . $row["GID"] . '</th> <th
align="center"> ' . $row["GardenerName"] . '</th> <th align="center"> '
. $row["ContactNo"] . '</th><th align="center"> ' . $row["Address"] . '</th><th
align="center"> ' . $row["DoE"] . '</th><th align="center"> '
. $dayofweek[$row["Holiday"]] . ' </th></tr>';
        }
        echo "</table>";
        echo "</div>";
    }
}

```

12) Listing Area Division Table from Database: -

```

$query = "SELECT * FROM Areas";

```

```

if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> Area Code
</th> <th align="center"> Area Name</th> <th align="center"> Location
</th></tr>';

        // output data of each row
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center"> <td
align="center">'. $row["AreaCode"]. '</td> <td
align="center">'. $row["AreaName"]. '</td> <td align="center">'
.$row["AreaLocation"]. ' </td></tr>';
        }
        echo "</table>";
        echo "</div>";
    }
}

```

13) Marking Attendance for Gardener : -

```

$query= "INSERT INTO Attendance VALUES ('$GID',CURRENT_DATE)";
if ($result = mysqli_query($db,$query)){
    echo "<script>alert('Successfully Marked!); window.location.href=
'./Attendance.php';</script>";
}

```

14) Monthly Attendance of Gardeners : -

```

$query = "SELECT GID,COUNT(At_Date) AS Num_of_Days FROM Attendance WHERE
MONTH(At_Date)='$SMonth' AND YEAR(At_Date)='$SYear' GROUP BY GID";

if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> GID
</th> <th align="center"> Number of Days </th></tr>';
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center"> <td align="center">'. $row["GID"]. '</td>
<td align="center">'. $row["Num_of_Days"]. '</td></tr>';
        }
        echo "</table>";
        echo "</div>";
    }
    else{
        echo "No Data Found.";
    }
}

```

```
}
```

15) Creating Weekly Duty Schedule: -

a) Getting Suggestions: -

```
$query = "SELECT GID FROM Gardeners WHERE Holiday!='$WDay' AND GID NOT IN (SELECT  
GID FROM DutyAssignment WHERE Wday='$d') LIMIT 5";  
  
if ($result = mysqli_query($db,$query)) {  
    if ($result->num_rows > 0) {  
        echo "<div class='form-div'>";  
        echo '<table border="1"><tr align="center"> <th align="center"> Suggested  
GIDs </th></tr>';  
        while($row = $result->fetch_assoc()) {  
            echo '<tr align="center"> <td  
align="center">'. $row["GID"]. '</td></tr>';  
        }  
        echo "</table>";  
        echo "</div>";  
    }  
    else{  
        echo "<div class='form-div'>";  
        echo "<p>No one is Available on this Day.<p>";  
        echo "</div>";  
    }  
}
```

b) Inserting new schedule in duty assignment table: -

```
$query= "INSERT INTO DutyAssignment VALUES ('$GID','$AreaCode','$d')";  
if ($result = mysqli_query($db,$query)){  
    echo "<script>alert('Successfully Assigned'); window.location.href=  
'./createSchedule.php';</script>";  
}
```

16) Viewing Weekly Duty Schedule for all Gardeners : -

```
$query = "SELECT * FROM DutyAssignment";  
$dayofweek = array("Sunday", "Monday", "Tuesday", "Wednesday", "Thursday",  
"Friday", "Saturday");  
if ($result = mysqli_query($db,$query)) {  
    if ($result->num_rows > 0) {  
        echo "<div class='form-div'>";  
        echo '<table border="1"><tr align="center"> <th align="center"> WeekDay  
</th> <th align="center"> Area Code </th> <th align="center"> GID </th></tr>';  
  
        // output data of each row  
        while($row = $result->fetch_assoc()) {
```

```

        echo '<tr align="center"> <td align="center">'
        . $dayofweek[$row["Wday"]]. ' </td> <td align="center">' . $row["AreaCode"] . '</td>'
        '<td align="center">' . $row["GID"] . ' </td></tr>';
    }
    echo "</table>";
    echo "</div>";
}
}

```

17) Viewing Equipments and tools stock :-

```

$query = "SELECT * FROM Equipments ";
if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> EID </th>'
        '<th align="center"> Equipment Name </th> <th align="center"> Area Code </th>'
        '</tr>';

        // output data of each row
        while($row = $result->fetch_assoc()) {
            echo '<tr align="center"> <td align="center">' . $row["EID"] . '</td>'
            '<td align="center">' . $row["EquipmentName"] . ' </td> <td'
            align="center">' . $row["Area"] . ' </td> </tr>';
        }
        echo "</table>";
        echo "</div>";
    }
}

```

18) Inserting new records in equipment database: -

```

$query= "INSERT INTO Equipments VALUES ('$EID','$AreaCode', '$EquipmentName')";

if ($result = mysqli_query($db,$query)){
    header("location: equipmentStock.php");
}

```

19) Viewing Maintenance Log for equipments: -

```

$query = "SELECT * FROM EqMaintenance ";
if ($result = mysqli_query($db,$query)) {
    if ($result->num_rows > 0) {
        echo "<div class='form-div'>";
        echo '<table border="1"><tr align="center"> <th align="center"> ReqID'
        '</th><th align="center"> EID </th> <th align="center"> MPlace </th> <th'
        align="center"> MContact </th> <th align="center"> MDate</th><th align="center">

```

```

MStatus </th></tr>';

    // output data of each row
    while($row = $result->fetch_assoc()) {
        echo '<tr align="center"><td align="center">' . $row["ReqID"] . '</td>
<td align="center">' . $row["EID"] . '</td> <td
align="center">' . $row["MPlace"] . '</td> <td align="center">'
        . $row["MContact"] . '</td> <td align="center">' . $row["MDate"] . '</td> <td
align="center">' . $row["MStatus"] . '</td> </tr>';
    }
    echo "</table>";
    echo "</div>";
}
}

```

20) Adding New entry in log for maintenance :-

```

$query= "INSERT INTO EqMaintenance (EID, MPlace, MContact, MDate, MStatus)
VALUES ('$EID', '$MPlace', '$MContact', CURRENT_DATE, 'Submitted')";

if ($result = mysqli_query($db,$query)){
    header("location: equipMaintenance.php");
}

```

21) Updating maintenance Status :-

```

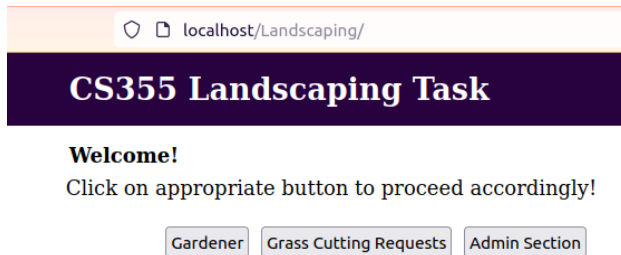
$query= "UPDATE EqMaintenance SET MStatus='$MStatus' WHERE ReqID='$ReqID'";
if ($result = mysqli_query($db,$query)){
    header("location: equipMaintenance.php");
}

```

4. Web Interface for this mini-project: -

I created the following pages to interact with the database designed above.

1. Homepage: This page takes you to either Administrator section, Gardener section or Grass cutting req portal.



localhost/Landscaping/

CS355 Landscaping Task

Welcome!
Click on appropriate button to proceed accordingly!

[Gardener](#) [Grass Cutting Requests](#) [Admin Section](#)

2. When we click on the Gardener button, we go into the gardener section where we can register and then login.



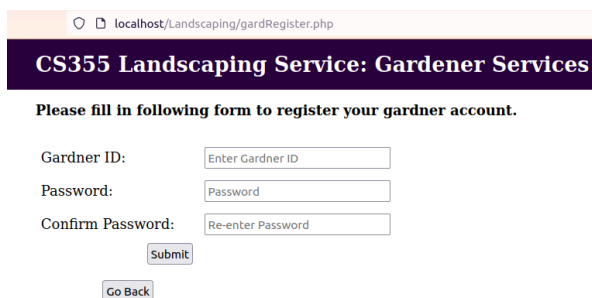
localhost/Landscaping/gardener.php

CS355 Landscaping Task : Gardener Services

Welcome!
Click on appropriate button to proceed accordingly!

[Login](#) [Register](#) [Go Back](#)

3. Clicking on the Register button sends us to the register page where we register gardnerID and create password for login.



localhost/Landscaping/gardRegister.php

CS355 Landscaping Service: Gardener Services

Please fill in following form to register your gardner account.

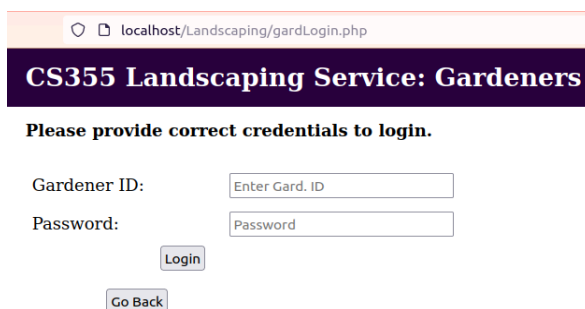
Gardner ID:

Password:

Confirm Password:

[Submit](#) [Go Back](#)

4. Let's fill it with gardenerID "BH001" and create a password and submit. We get redirected to the login page if registration is successful.



localhost/Landscaping/gardLogin.php

CS355 Landscaping Service: Gardeners

Please provide correct credentials to login.

Gardener ID:

Password:

[Login](#) [Go Back](#)

5. After filling "BH001" and the password we created we can login into the gardener's profile page.

localhost/Landscaping/gardProfile.php

CS355 Landscaping Service : Gardener Details

Welcome BH001
Your profile details are as follows:

Gardener ID: BH001
 Name: Saurabh
 DoE: 2020-10-01
 Holiday: Monday
 Address: Bihta
 Contact No.: 8765640760

6. We can click on update contact number to update gardener's contact no, clicking on Area division shows us how campus is divided into various areas and view schedule displays Gardener's work schedule.

localhost/Landscaping/gardenerProfileUpdate.php

CS355 Landscaping Services : Gardeners

Welcome BH001
Please fill out all fields with updated information.

Mobile No:

Schedule:

localhost/Landscaping/gardSchedule.php

CS355 Landscaping Service: Gardeners

Welcome BH001
Your Schedule is as follows:

Day	Area
Sunday	NewBoysHostel
Tuesday	NewBoysHostel
Wednesday	NewBoysHostel
Thursday	NewBoysHostel
Friday	NewBoysHostel
Saturday	NewBoysHostel

Campus Areas:

localhost/Landscaping/viewAreas.php

CS355 Landscaping Service: Area Division

Welcome!

Campus is divided into areas are as follows:

Area Code	Area Name	Location
1	Block4	Academic Section
2	Block1	Academic Section
3	Block2	Academic Section
4	Kalam Hostel	Students Hostel Area
5	Block3	Academic Section
6	Football Ground	Sports Area
7	Admin Building	Administrative Section
8	Workshop	Department Workshops
9	Tutorial Block	Academic Section
10	Building1	Faculty Residential Area
11	Cricket Ground	Sports Area
12	NewBoysHostel	Students Hostel Area
13	Gymkhana	Gymkhana Building
14	Building3	Faculty Residential Area
15	Block6	Academic Section

7. Lets log out and check out grass cutting request system : -

localhost/Landscaping/viewGCReq.php

CS355 Landscaping Service: Gardening Requests

Welcome!

Current Requests are as follows:

ReqID	Area Code	ReqEmpID	ReqContact	ReqStatus
1	12	1901EE44	7355409262	Finished
2	1	1901CS77	6756876756	Finished
3	2	1901CS75	6027364523	Finished
4	1	1901CS77	7656098902	Submitted
5	2	1901EE68	7979004304	Submitted

[Area Division](#) [Submit New Req](#) [Go Back](#)

8. We can use submit New Req button to submit a new request.

localhost/Landscaping/GCReq.php

Landscaping Service : Grass Cutting Requests

Please fill in following form to submit grass cutting requests for any specific area.

Employee ID:

Area Code:

Contact No:

Lets submit a req by empID: 1901CS77 for AreaCode 5 with a random contactNo.

9. After successful submission we are redirected to request's section where we can see our new request.

localhost/Landscaping/viewGCReq.php

CS355 Landscaping Service: Gardening Requests

Welcome!
Current Requests are as follows:

ReqID	Area Code	ReqEmpID	ReqContact	ReqStatus
1	12	1901EE44	7355409262	Finished
2	1	1901CS77	6756876756	Finished
3	2	1901CS75	6027364523	Finished
4	1	1901CS77	7656098902	Submitted
5	2	1901EE68	7979004304	Submitted
6	5	1901CS77	7867568765	Submitted

[Area Division](#)
[Submit New Req](#)
[Go Back](#)

Administrators can login and update status once request is fulfilled.

10. Now coming back to the homepage when we click for Admin Section, we can register for an admin account or login, Registration only generates a username and password but we cannot use it to login unless it is authorised by another Admin.

Admin Section Homepage: -

localhost/Landscaping/admin.php

CS355 Landscaping Task : Administrative Section

Welcome!
Click on appropriate button to proceed accordingly!

[Login](#)
[Register](#)
[Go Back](#)

Admin account register form: -

localhost/Landscaping/adminRegister.php

Landscaping Service : Administrative Services

Please fill in following form to register your Admin account.

Username:

Password:

Confirm Password:

[Submit](#)

[Go Back](#)

After filling in and once our account is authorised we can login using this username password. For further tasks let's use an already authorised account username "abhay376" and password "12345678".

Login Page: -

localhost/Landscaping/adminLogin.php

CS355 Landscaping Service: Admin Section

Please provide correct credentials to login.

Username:

Password:

[Login](#)

[Go Back](#)

After successful login we are sent to our welcome page where we get various functionalities.

localhost/Landscaping/adminWelcome.php

CS355 Landscaping Service: Admin Homepage

Welcome abhay376
Please click on following buttons to continue:

Make an Admin Area Division Gardeners Details View/Update Equipment Stock Attendance Create/View Schedule View Grass Cutting Requests Log out

i) Make an admin: this button opens up a form where we can fill in a username to authorise admin access.

localhost/Landscaping/makeAdmin.php

CS355 Landscaping Services: Admin

Welcome abhay376
Fill the form below to make a user Admin:

Username:

Make Admin

Go Back

ii)Gardener Details: This button lists out all gardener's details currently employed.

localhost/Landscaping/gardenerList.php

CS355 Landscaping Service: Gardener List

Welcome!
List of Currently employed Gardeners is as follows:

ID	GardenerName	ContactNo	Address	DoE	Holiday
BH001	Saurabh	8765640765	Bihta	2020-10-01	Monday
BH002	Ankit	9878675676	Danapur	2020-01-07	Thursday
BH003	Shubham	9878675645	Danapur	2021-08-10	Sunday
BH004	Sumit	7867564576	Bihta	2021-10-07	Thursday
BH005	Sanjay	7989786756	Danapur	2021-09-10	Saturday
BH006	Kumar	6356768798	Danapur	2021-10-10	Wednesday
BH007	Kuldeep	9946578656	Bihta	2019-06-07	Sunday
DC001	Gaurav	7656476534	Bihta	2021-04-10	Saturday
DC002	Ramesh	8767567687	Bihta	2021-10-07	Tuesday
DC003	Suman	7867564576	Bihta	2021-06-19	Monday
DC004	Pankaj	6787876798	Bihta	2021-11-10	Friday
DN001	Dhruv	7658745767	Bihta	2021-09-05	Wednesday
DN002	Shankar	8767567525	Danapur	2021-05-07	Friday
DN003	Vishal	7989564565	Bihta	2021-11-01	Sunday
DN004	Shushil	7987675623	Bihta	2021-04-16	Tuesday

Insert New Gardener Go Back

iii) clicking Insert new gardener opens up a form where we can fill in another record for any new gardener in the database.

localhost/Landscaping/insertGardener.php

CS355 Landscaping Service: Gardeners

Welcome abhay376
Fill the form below to insert New Gardener:

GID:

Gardener Name:

Address:

Holiday:

ContactNo:

DoE:

iv) another button on adminwelcome page is Attendance, where we can mark attendance for any given GID. We cannot choose date for security purposes, it automatically takes in the current server date.

localhost/Landscaping/attendance.php

CS355 Admin Service: Attendance

Welcome abhay376
Fill the form below to update Attendance:

GID:

From here we can also see our monthly attendance for any specific month.

localhost/Landscaping/monthlyAttendance.php?Year=2021&Month=11

CS355 Landscaping Service: Monthly Attendance

Welcome!
Monthly Attendance of all Gardeners:

Year:

Month:

GID	Number of Days
BH001	5
BH002	5
BH003	4
DC001	6
DC002	5
DN001	4
DN002	6

v) Next button on the homepage is for maintenance of equipment records.

When we click on it, we can see the current stock of equipment and tools and we also get a form to add a new one.

CS355 Landscaping Service: Equipments and Tools

Welcome abhay376

Current Equipment Stock is as follows:

EID	Equipment Name	Area Code
CS104	Chainshaw	1
FK001	Fork	1
FK002	Fork	1
FK003	Fork	3
FK004	Fork	4
FK005	Fork	3
FK006	Fork	6
LM101	LawnMover	1
LM102	LawnMover	12
LP103	Loppers	2
PS001	Pruners	3
RK001	Rake	2
RK002	Rake	12
SH101	Shovel	1
SH103	Shovel	12
TW001	Trowel	6

EID:

Area Code:

Equipment Name:

vi) After clicking view maintenance status, we can also see the current maintenance log, insert a new value or update status of old log.

CS355 Landscaping Service: Equipments and Tools

Welcome abhay376

Current Equipment Maintenance Log is as follows:

ReqID	EID	MPlace	MContact	MDate	MStatus
1	SH101	Bihta	6756453464	2021-10-10	Finished
2	LM102	Patna	6756453465	2021-10-28	Finished
3	LM101	Patna	7656453423	2021-11-27	Sent
4	SH103	Bihta	9878675647	2021-11-28	Finished
5	FK001	Bihta	6545876565	2021-11-28	Sent

EID:

Maintainer Place:

Contact No:

Status Update Form: -



localhost/Landscaping/updateMStatus.php

CS355 Landscaping Service: Equipments and Tools

Welcome abhay376

Fill the form below to update maintenance status:

ReqID:

Maintenance Status:

vii) Similarly heading to next section Grass Cutting req, as an Admin we get to see a new button to update status of req if it has been fulfilled.



localhost/Landscaping/updateReqStatus.php

CS355 Landscaping Service: GC Requests

Welcome abhay376

Fill the form below to update GrassCutting Request status:

ReqID:

Request Status:

viii) Last button other than logout in our Admin homepage is for schedule.

Clicking this button opens up a form where we can fill our desired day to get a list of available gardeners for scheduling on that day and we can assign them any area code of our choice.

localhost/Landscaping/createSchedule.php?WDay=4

CS355 Landscaping Service: Assign Duty

Welcome!
Fill the following form to assign Duty to a Gardener

Day:

Suggested GIDs
BH003
BH005
BH006
BH007
DC002

Day:

Area Code:

Gardener:

Here let's assign Area 5 to BH003 on day4 i.e. Wednesday.
 Then we can click on current schedule to see updated Schedule: -

localhost/Landscaping/viewSchedule.php

CS355 Landscaping Service: View Duty Assignments

Welcome!
Current Schedule is as follows:

WeekDay	Area Code	GID
Monday	1	BH003
Sunday	1	DC001
Wednesday	1	DC001
Tuesday	1	DC003
Thursday	1	DN001
Saturday	1	DN001
Thursday	1	DN002
Friday	1	DN002
Sunday	2	BH005
Wednesday	5	BH003
Sunday	12	BH001
Tuesday	12	BH001
Wednesday	12	BH001
Thursday	12	BH001
Friday	12	BH001
Saturday	12	BH001
Monday	12	BH002
Sunday	12	DC002

viii) After Logout we return to our mainpage index.php

All Php Files, CSV Data used to fill the tables in Landscaping Task can be seen at:-
[Task3 SourceCode](#)